

The grammar of gender:

Insights from Bantu asymmetries of AGR with conjoined subjects

Vicki Carstens
University of Connecticut

1. Introduction

1.1 Overview

Much recent morpho-syntactic theory analyzes genders as having a mixture of *i*(nterpretable) and *u*(ninterpretable) content (Kramer 2009, Percus 2011, Matushansky 2013). In this way predictable mappings, like that of female and male-denoting nouns to feminine and masculine genders respectively, are reconciled with the existence of arbitrary members, hypothesized to bear *u*versions of the same genders. Attributing gender features to “little *n*” categorizers yields the analyses in (2) for (1)a-c (adapted from Kramer 2015: 91-96).

- | | | |
|--|--|--|
| (1) a. la mujer
the.FEM woman
‘the man’ | b. el hombre
the.MASC man
‘the woman’ | c. la pluma [Spanish]
the.FEM pen
‘the pen’ |
| (2) a. [_{NP} [_{VP} MUJER] <i>n</i> _[ifem]] | b. [_{NP} [_{VP} HOMBRE] <i>n</i> _{i[-fem]}] | c. [_{NP} [_{VP} PLUMA] <i>n</i> _[ufem]] |

Associations between genders and meanings are less clear in a Bantu language like Xhosa, which has five singular/plural noun class pairings exemplified in (3).¹ Starting with Watkins (1937) various researchers have argued that genders underlie each pair of Bantu classes. But there is considerable arbitrariness to class memberships, engendering skepticism about their having semantic content (see Carstens 2008, 2011, Fuchs & van der Wal 2018, Maho 1999,

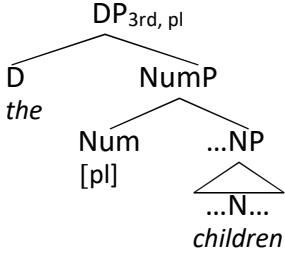
¹Examples are from my own fieldwork unless otherwise noted. In glosses, numerals indicate noun class (number+gender). Classes 1 vs. 1a, 2 vs. 2a differ only in their nominal prefixes (cl. 1 *um-* vs. 1a *u-*; cl. 1 *aba-* vs. 1a *oo-*) 1a/2a include kinship terms, proper names, and arbitrary members of 1/2. Singular/plural class pairings share color-coding, to aid recognition. SM = subject marker; OM = object marker; ASP = aspect; DISJ = disjoint, a VP-final verb form; LOC = locative. PST = past tense; PRT = participle; FEM = feminine; MASC = masculine; NEUT = neuter; PL = plural; POSS= possessor. [X] denotes semantic feature X. Vowel coalescence affects the CV conjoiner *na* – ‘and’ and the vocalic outer class marker (aka augment) of the following noun, to which it prefixes. Empirical generalizations discussed here came to my attention in Taraldsen et al (2018), reporting on fill-in-the-blank subject marker questionnaires completed by 21 Xhosa-speaking students at Stellenbosch University in 2013 and 2014. I administered fill-in-the-blank questionnaires to 15 Xhosa-speaking students at University of the Western Cape and Stellenbosch University in 2025. See §1.2 for details.

among others on the unreliability of semantic characterizations for most Bantu noun classes).

- (3) a. **um**-ntu/**aba**-ntu b. **um**-pu/**imi**-pu c. **ili**-so/**ame**-hlo d. **isi**-tya/**izi**-tya e. **in**-ja/**izin**-ja
 1-/2-person 3-/4-gun 5-/6-eye 7-/8-dish 9-/10-dog
 'person/s' 'gun/s' 'eye/s' 'dish/es' 'dog/s'

This paper provides evidence from complex patterns of agreement with conjoined singulars that the Xhosa class system partitions nouns among three core genders with the semantic features *i*[human], *i*[animal], and *i*[inanimate], though this is camouflaged by rather free stacking of one *nP* above another.² Patterns of plural agreement with conjoined singulars are a key diagnostic for the presence or absence of interpretable content within a gender, I will argue, motivating this proposal. Moreover, close consideration of the Xhosa facts yields important insights into the mechanics of how agreement with conjuncts works.

The analysis relies on the Agree operation of Chomsky (2000, 2001), illustrated below for subject agreement with a plural DP. (4)a illustrates a prevalent view that Num(ber) heads a projection within the DP's middle field; Num-raising or percolation shares the singular or plural feature with DP (among others see Ritter 1991, Danon 2011; Carstens 1991, 2011). Agreeing *are* in (4)b results from Agree applying between unvalued phi-features (*uPhi*) of T(ense) and the DP *the children* as shown (4)c (strike-through indicates successful *uPhi* valuation).

- (4) a. 
- b. The children are smart.
- c. Agree (T_{*uPhi*}+be, DP_{3,pl}) → ~~be_{*uPhi*3,pl}~~ = *are*

² See also Halpert & Hammerly (2025) who propose the same idea in simultaneous independent work.

In contrast, a conjunction of singulars like (5) would seem to value plural agreement based on the sum of the conjuncts' features in what is generally called *resolved agreement*.

(5) [DP.sg the child] and [DP.sg the adult] *are*/**is* smart.

Grammatical gender adds a complicating feature to resolved agreement, absent in a language like English. Patterns of agreement with gendered conjuncts are a rich source of information regarding how resolution comes about.

In Xhosa there are two main patterns of agreement with conjoined singulars whose gender features match, henceforth uniform conjuncts, depending on the genders involved.

Uniform conjuncts drawn from three of Xhosa's five genders generally pair with gender-matching plural agreement, regardless of the semantic properties of the conjoined nouns. This is the case in Xhosa (6)a,b where the class 2 subject marker accompanying conjoined class 1 nouns is the same as accompanies a plural subject in (6)c; in (7)a-c we see the same pattern of expected class 8 plural agreement with conjoined class 7 nouns. One further pair of classes, 9/10, patterns with 1/2 and 7/8 in this respect.

- (6) a. Um-mi nom-ongameli **ba**-ya-ncokola. [1&1=2]
 1-citizen and.1-president 2SM-DISJ-chat
 'The citizen and the president are chatting.'
- b. U-L no-M **ba**-phezu kwetafile. [1&1=2]
 1a-L and.1a-M 2SM-are.on LOC.table
 'The L and the M are on the table.'
- c. Aba-mi/oo-bhaka **b**-a-wa.
 2-citizens/2a-backpacks 2SM-ASP-fall
 'The citizens/backpacks fell.'
- (7) a. Isi-bane nesi-tya **zi**-nyamalele. [7&7=8]
 7-lamp and.7-dish 8SM-disappeared
 'The lamp and the dish have disappeared.'
- b. Is-anuse nes-azi **zi**-ya-sebenza.
 7-diviner and.7-scientist 8SM-DISJ-work
 'The diviner and the scientist are working.'
- c. Izi-bane/iz-azi **zi**-nyamalele.
 8-lamps/8-scientists 8SM-disappeared
 'The lamps/scientists disappeared.'

These facts are compatible with a simple approach to conjunct agreement, adding up the DPs'

The second pattern introduces intriguing complications, however: for uniform conjuncts of two other genders including class 5 below, gender-matching plural agreement is unacceptable as noted in Taraldsen et al (2018), Voeltz (1971), Roberts & Wolontis (1974) (and see Corbett & Mtenje 1987 for very similar facts in Chichewa). Instead, some other agreement is employed, most often class 2 *ba-* for human-denoting conjuncts like (8)a, the same as for conjoined class 1 singulars and class 2 plurals like (6)a-c. For the inanimates in (8)b the subject marker is class 8 *zi-*. (8)c,d show that the class 6 *a-* subject marker, ruled out in (8)a,b, is required with plural subjects of this gender. Xhosa class 3 nouns exhibit a parallel pattern.

- The examples in (9) show that conjoined singulars of mismatched genders also pair with class 2 if human-denoting and class 8 if inanimate. Thus mismatched singular conjunctions on the one hand and uniform conjunctions of classes [5+5] and [3+3] on the other pattern alike.

- I argue that these patterns reveal several properties of gender semantics and their role in the mechanics of resolved agreement, in Xhosa and more broadly.

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denoting nouns in (8)a and (9)a are layered above cores of [human] gender 1/2 as illustrated in (10). Similarly, inanimate nouns like those of (8)b and (9)b have cores of [inanimate] 7/8.

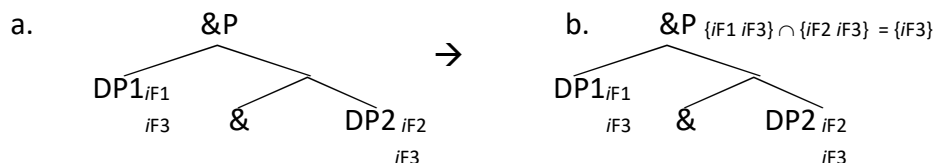
Stacking explains the scattering of semantic fields across the Xhosa noun classes; see also Fuchs & Van der Wal (2022) for a stacking approach to Bantu diminutives and locatives.

(10) Sample structures of [human] nouns: a core of interpretably [human] 1/2

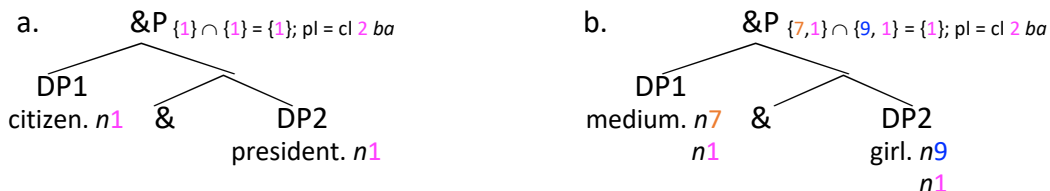
- a. um-ntu/aba-ntu [in_{1/2} VMNTU] in_{1/2} for [human]
 1-person/2-people
 'person/s'
- b. i-gqwetha /ama-gqwetha [un_{5/6} [in_{1/2} VGWETHA]] un of 5/6 stacks above 1/2 in
 5-lawyer/6-lawyers
 'lawyer/s'

The second pillar is a proposal of Adamson & Anagnostopoulou (2024; henceforth A&A) that resolved agreement is based on *percolation* of conjuncts' features to their containing Conjunct Phrase (&P) and *conversion* to form a single set, based on the *intersection* of the conjuncts *interpretable* features as shown in (11). (12)a,b sketch how this works to derive agreement with matched and mismatched conjuncts (6)a and (9)a respectively.^{3,4}

(11) *Feature percolation and conversion via intersection of igender features*



(12) *Agreement with Xhosa conjoined singulars via intersection*

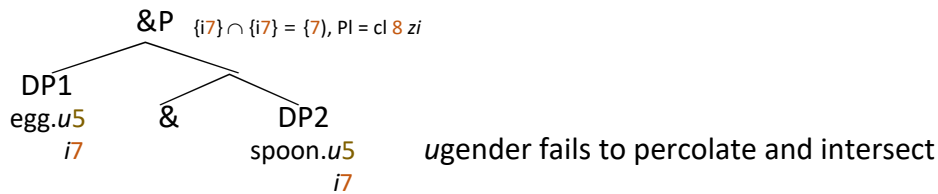


³ I leave aside temporarily the *u-* vs. *i-* status of the higher gender features in (12)b.

⁴ An anonymous reviewer points out that Dalrymple & Kaplan (2000), Sadler (2003) and Kuhn & Sadler (2007) among others offer detailed models of feature-resolution in the LFG framework as does Moosally (1998) on Zulu. While these considerably predate the approach of Adamson & Anagnostopoulou that I adopt, length considerations preclude discussion and comparison.

I take the failure of [5&5], [3&3] to pair with gender-matching plural agreement as evidence that these genders are strictly uninterpretable; their features never populate the intersection.

(13) *class 8 agr with conjoined inanimates* [5+5] in (8)b:



Now notice that in contrast to Xhosa [3&3], [5&5], the gender features of arbitrary conjuncts [1&1], [7&7] in (6)b and (7)b *do* pair with matching agreement. The approach I have sketched to the Bantu agreement asymmetries for uniform conjoined singulars is incompatible with the view that arbitrary members bear *u*gender since this would predict matching agreement with them to fail just as it does for [5&5] in (8)b/(13). Therefore in a departure from the approach sketched in (2) I argue that genders with *i*content are always semantic to some degree, encoding *i*[entity] for otherwise arbitrary members like those in (6)b and (7)b. Their percolated genders can be intersected as in (11) and (12), and value agreement.

In Xhosa, the genders in which matching plural agreement is successful with uniform conjuncts outnumber those in which it fails by 3:2. In lieu of (13) it is conceivable that some special featural idiosyncrasies of Xhosa’s “outlier” genders 3/4 and 5/6 might underlie the pattern in (8)b, as in existing analyses of a similar agreement failure with paired neuters in BCS ((14) from Despić 2016, and see also Nevins 2018, Tsimplici & Hulk 2013, Arsenijević 2017, 2021).

(14) Sel-o i polj-e su poltopljen-i /*poltopljen-a. [BCS]
 village-NEUT.SG and field.NEUT.SG AUX.PL flooded-✓MASC.PL/ *NEUT-PL
 ‘The village and the field are flooded.’

But I will show that diagnostics for any relevant idiosyncrasies fail for Xhosa 3/4 and 5/6.

Moreover, in related Shona, where there are eight singular/plural noun class pairings, the

pattern is reversed: those genders in which matching agreement is impossible with uniform conjoined singulars outnumber those where it succeeds by 6:2 — agreement with conjoined singulars is *always* class 2 for humans and 8 for non-humans. Thus genders in which matching agreement *succeeds* are the outliers. Class 9 is representative: (15)a demonstrates that its plurals are found in class 10; (15)b,c show that conjoined singulars cannot pair with 10.

- (15) a. Im-bwa/n-herera **dz**-aka-on-ana. [Shona: 10agr = *dzi*-]
 10-dogs/10-orphans 10_{SM}-PST-see-RECIP
 'The dogs/orphans saw each other'
- b. N-herera ne-n-yanzvi **va**- /***dz**-aka-on-ana. [Shona: 9&9=2]
 9-orphan and-9-expert 2_{SM}-/*10_{SM}-PST-see-RECIP
 'The orphan and the expert saw each other.'
- c. Im-bwa nhema ne-m-hou chena **zv**- /***dz**-aka-rwa. [Shona: 9&9=8]
 9-dog 9-black and-9-cow 9-white 8_{SM}-/*10_{SM}-PST-fall
 'The black dog and the white cow fell.'

I propose that no Shona genders apart from 1/2 and 7/8 have associated semantic content, which is crucial for matching plural agreement under this approach. Generalizing, what is common, cross-linguistically, to genders in which matching plural agreement succeeds with uniform conjuncts is semantic association to a salient class of entities, absent in genders where such matching agreement fails. A small number of genders which exhibit such associations within a given language have privileged status as interpretable genders, with corresponding legibility in the morphosyntax of conjunct agreement. One consequence is that while a semantic core is arguably criterial to gender systems (Kramer 2015:70), not all genders in a given language necessarily have one. Skepticism about semantic content for *some* Bantu classes is validated, but the upshot is that both perspectives on this question are partly correct.

Summing up, this paper finds striking confirmation in Xhosa for the *n* and *nP*-stacking approach to gender, for the *u*- vs. *i*- distinction in nominal genders, and for semantic cores

underlying the Bantu noun class system and coexisting with much arbitrariness. I argue that *nP* stacking explains two near-contradictory patterns: on the one hand, the dispersal of semantic content across Xhosa classes and on the other hand, the semantic consistencies of default agreement. My analysis provides a unified account of these through a minimal adaptation of Kramer (2015) and Adamson & Anagnostopoulou (2024), coupled with an innovative approach to arbitrary members of genders with semantic content.

1.2 Methodology

As mentioned in note 1, the empirical baseline for this paper is Taraldsen et al (2018)'s report on results from fill-in-the-missing-subject-marker questionnaires administered to twenty-one Xhosa-speaking students at Stellenbosch University in 2013 and 2014. I have conducted research administering the same type of fill-in-the-blank questionnaires to fifteen Xhosa speaking students at University of the Western Cape (UWC) and Stellenbosch University, and three Zimbabwean Shona speakers (one based in New Haven by email in and two in person at Stellenbosch Institute for Advanced Study). Places of origin of the Xhosa speakers in both my studies and Taraldsen et al's varied across the Eastern Cape and the Cape Town area; this is difficult to avoid when participants are college students drawn from different home locations.

My questionnaires were developed with the help of two Xhosa speakers at UWC, one in Stellenbosch, and the New Haven-based Shona speaker mentioned above. They included at least two conjunctions each for [human & human] and an assortment of non-human conjuncts including [inanimate & inanimate], [animal & animal] and [tree & tree] within pairings [1&1], [1&3], [1&5], [3&1], [3&3] etc. Xhosa questionnaires were mainly administered in person, with myself and a speaker facing my computer screen; the speakers read aloud each example,

verbally adding their choices of subject marker, and I typed these into the gaps under their supervision. I posed follow-up questions by email and in online meetings.

1.3 Structure of the paper

This paper has seven sections. §2 presents the basics of Xhosa and Shona noun classes and agreement. §3 details the asymmetries in agreement with conjoined singulars in Xhosa. §4 and §5 lay out the analysis. §6 explores and rejects an alternative possibility: that non-canonical semantic content makes Xhosa 3/4 and 5/6 defective, applying diagnostics from Arsenijević (2017, 2021)’s work on BCS neuter with negative results. §7 concludes.

2. Xhosa and Shona basics

2.1 Noun class overview

As noted in §1, Xhosa has five regular singular/plural pairings exemplified in (3) (repeated below). These pairings are generally identified as noun classes 1-10 following a numbering system that dates to Meinhof (1906) and Guthrie (1948, 1971).

- (3) a. **um**-ntu/**aba**-ntu b. **um**-pu/**imi**-pu c. **ili**-so/**ame**-hlo d. **isi**-tya/**izi**-tya e. **in**-ja/**izin**-ja
 1-/2-person 3-/4-gun 5-/6-eye 7-/8-dish 9-/10-dog
 ‘person/s’ ‘gun/s’ ‘eye/s’ ‘dish/es’ ‘dog/s’

Shona has eight such pairings exemplified in (16).⁵

- (16) a. **mu**-/**va**-sikana b. **mu**-/**mi**-romo c. **zai**/**ma**-zai d. **chi**-/**zvi**-ngwa
 1/2-girl 3/4-finger 5/6-egg 7/8-bread
 ‘girl/s’ ‘finger/s’ ‘egg/s’ ‘loaf/loaves of bread’
 e. **im**-/**im**-bwa f. **ru**-kova/hova g. **u**-/**ma**-uta h. ka-/tu-sikana
 9/10-dog 11/10-stream 14/6-bow **12/13**-girl
 ‘dog/s’ ‘stream/s’ ‘bow/s’ ‘tiny girl/s’

A number of researchers have argued that grammatical genders underlie each Bantu

⁵ Both languages have a few classes that do not pluralize; these won’t be relevant here. Plurals of Shona classes 11 and 14 are homophonous with those of 9 and 5 respectively; Bantuist linguistic convention treats them as classes 10 and 6 (see §3.5), reflecting proto-Bantu reconstructions to appear in Table 1.

singular/plural noun class pairing (see Carstens 1991, 2008, Corbett 1991, Corbett & Mtenje 1987, Fuchs & van der Wal 2022, Watkins 1937).⁶ Example (17) shows the system of five genders A-E proposed in Carstens (1991) for noun classes 1-10. Carstens proposed that Bantu nouns raise and incorporate to Num, and noun class prefixes spell out number features in a gender-particular way as shown in (18).⁷ Updating this to reflect the *n* approach to gender and an Agree relation feeding both incorporation (Roberts 2019) and the gendered expression of number yields (19), where $u\phi$ of Num probes and Agrees with $n(P)$.

(17) Bantu Genders (Carstens 1991)

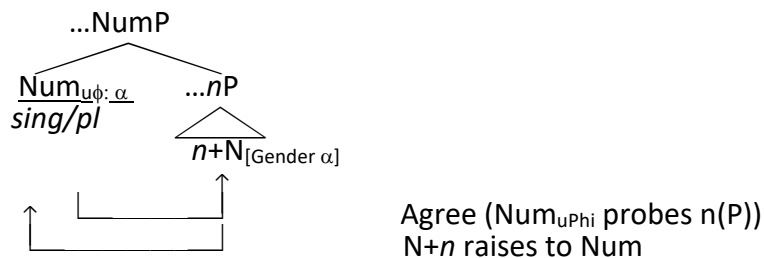
Gender **A**: stems of Classes 1/2
 Gender **B**: stems of Classes 3/4
 Gender **C**: stems of Classes 5/6

Gender **D**: stems of Classes 7/8
 Gender **E**: stems of Classes 9/10
 Etc.

(18) Nouns class prefixes as gender-specific number morphology (Carstens 1991, 2008)

[Singular] $\longleftrightarrow$ **um-** / __N_[Gender A] [Plural] $\longleftrightarrow$ **aba-** / __N_[Gender A]
 [Singular] $\longleftrightarrow$ **isi-** / __N_[Gender D] [Plural] $\longleftrightarrow$ **izi-** / __N_[Gender D] etc.

(19)



I adopt Carstens's analysis of noun classes as singulars and plurals of genders A-E, but for

⁶ I will support the gender approach to class with two diagnostics, agreement and a semantic core. However, as an anonymous reviewer points out, noun class is a complex and controversial topic. Many researchers have sought to reconcile the fact that within Bantu classes, recognizable categories of entities coincide with apparently arbitrary members and with recurring properties like shape and size. Some have proposed abstract meaning networks (Contini-Morava 2000, Dechaine et al 2014, Hendrikse 2001, Selvik 2001); others sought to generalize from shifts in noun class that accompany derivational processes, proposing a derivational analysis for the singular/plural function of class (Schadeberg 2001 among others). Marten (2021) provides an overview and references to relevant literature outside the scope of this brief introduction.

⁷ There are minor variations in class prefix forms. In Xhosa, some nouns of 9/10 including borrowings bear *i-*/*ii-* prefixes though the canonical prefixes for 9/10 native nouns are *in-*/*iin-*. Other variations include nasal and vowel assimilation, some prefixal vowel deletions, the difference already mentioned between 1/2 *um-*/*aba-* and 1a/2a *u-*/*oo-*, and a minimum word size effect: in class 10, monosyllabic noun stems bear *izi(n)-* as opposed to the standard class 10 *ii(n)-* prefixes, and some] class 5 monosyllabics bear *ili-* rather than *i-*.

convenience I utilize the traditional numbering system in glosses and much of the discussion.

The gender analysis of Bantu noun class follows from standard definitions of grammatical gender, which have changed little over the decades. (20) is a synthesis from Kramer (2015):⁸

(20) Gender is ... the sorting of nouns into two or more classes as reflected in agreement on articles, adjectives, verbs, and other syntactic categories (Kramer 2015:1, citing Hockett 1958:231, Fodor 1959:3, Corbett 1991:1, Comrie 1999:457, Matosovic 2004: 19-20).

Xhosa and Shona's abundant agreement is exemplified in (21) and (22).

- (21) a. **Laa** **m**-ntu **um**-khulu **u**-zo-wa. b. **Elo** **li**-tye **eli**-khulu **li**-zo-wa. [Xhosa]
 1-that 1-person 1-big 1SM-FUT-fall 5-that 5-stone 5-big 5SM-FUT-fall
 'That big person will fall.' 'That big stone will fall.'
- (22) a. **Mu**-rume **mu**-kuru **iyeye** **a**-cha-wa. b. **I**-bwe **Ø**-guru **iroro** **ri**-cha-wa. [Shona]
 1-man 1-big 1.that 1SM-FUT-fall 5-stone 5-big 5.that 5SM-FUT-fall
 'That big man will fall.' 'That big stone will fall.'

Agreement with non-conjoined DPs in Xhosa is only based on class, not semantic features. Thus

'woman' and 'train' of (23)a and (24)a control the same agreements since both are class 1(a).

Likewise for Xhosa class 9 nouns 'girl' and 'mail' of (23)b, (24)b and class 5 'lawyer' and 'spoon'

of (23)c, (24)c; the plurals in (25) follow suit. We'll see that class 1 is the canonical class of

singular humans, yet class 1 agreement with class 9 'girl' and class 5 'lawyer' is not possible.⁹

- (23) a. **um**-fazi **w**-a-fika. b. **in**-tombi **y**-a-fika. c. **i**-gqwetha **l**-a-hleka.
 1-woman 1SM-PST-arrive 9-girl 9SM-PST-arrive 5-lawyer 5SM-PST-laugh
 'The woman arrived.' 'The girl arrived.' 'The lawyer laughed.'
- (24) a. **u**-lolilwe **w**-a-fika. b. **i**-posi **y**-a-fika. c. **i**-cepe **l**-a-wa.
 1a-train 1SM-PST-arrive 9-mail 9SM-PST-arrive 5-spoon 5SM-PST-fall
 'The train arrived.' 'The mail arrived.' 'The spoon fell.'
- (25) a. **Aba**-fazi/**oo**-lolilwe **ba**-a-fika. b. **lin**-tombi/**iin**-cwadi **z**-a-fika
 2-women/2a-trains 2SM-PST-arrive 10-girls/10-letters 10SM-PST-arrive
 'The women/trains arrived.' 'The girls/letters arrived.'
- c. **Ama**-gqwetha/**ama**-tye **a**-a-wa. d. **Imi**-thi/**imi**-gewu **y**-a-wa.
 6-lawyers /6-stones 6SM-PST-fell 4-trees/4-criminals 4SM-PST-fell
 'The lawyers/stones fell.' 'The trees/criminals fell down.'

⁸ As a reviewer points out, Kramer 2015:70 adds as criterial the existence of a semantic core; see §2.2.1.

⁹ A reviewer asks if there is hybrid agreement such that e.g. some [human] nouns can alternatively pair with semantic agreement. This has not been described and is not accepted by my Xhosa or Shona consultants.

Examples (26)-(27) illustrate similar phenomena for Shona non-conjoined DPs. A noun of canonically human classes 1/2 necessarily pairs with class 1 *a-* and class 2 *va-* agreement in (26)a,b. Class 9 ‘bowl’ and ‘orphan’ both pair with *i-* agreement (contracted to *y-* before a vowel) in (27)a-b; the plurals of both nouns can pair only with class 10 agreement as in (27)c.

- (26) a. Mw-ana a-aka-nyemwerera. b. V-ana v-aka-nyemwerera. [Shona]
 1-child 1SM-PST-smile 2-child 2SM-PST-smile’
 ‘The child smiled.’ ‘The children smiled.’
- (27) a. Mbiya y-aka-tsemuka. b. N-herera y-aka-nyemwerera.
 9-bowl 9SM-broke 9-orphan 9SM-PST-smile
 ‘The bowl broke.’ ‘The orphan smiled.’
- c. N-herera /M-biya dz-aka-donha.
 10.orphans/10.bowls 10SM-PST-fall
 ‘The orphans/bowls fell.’

In the interests of brevity §2.2-§3.4 exemplify phenomena of interest with representative Xhosa data. I return to Shona in §3.5, demonstrating that while its noun class and agreement phenomena parallel those of Xhosa, facts of coordinated singulars drawn from its larger number of genders yield a novel insight crucial to the construction of a unified theory.

2.2 Semantics of noun classes

2.2.1 Proto-Bantu

Gender systems typically exhibit strands of semantic unity associated with at least some of their contents; Kramer 2015:70 proposes that like agreement, a semantic core is a criterial property of gender systems (see also Adamson & Anagnostopoulou 2024, henceforth A&A). Consistent with this, reconstructions of Proto-Bantu noun classes have associated meanings including [human], [animal], [tree], [diminutive], and locations ‘in’, ‘on’, and ‘at’. Table 1 is fairly standard in this regard (see Maho 1999 for detailed discussion of reconstructed class contents).

Table 1: A Proto-Bantu noun class reconstruction (Good 2020:151)

Singular (number) ♦	Singular (form) ♦	Plural (number) ♦	Plural (form) ♦	Semantics ♦
1	*mù-	2	*βà-	humans
3	*mù-	4	*mì-	trees, plants
5	*lì-	6	*mà-	mixed/cl. 6 liquids
7	*kì-	8	*βì-	mixed
9	*nì-	10	*lì-nì-	animals, mixed
11	*lù-			mixed
12	*kà-	13	*tù-	augmentative, diminutive, etc.
14	*βù-			abstract
15	*kù-			infinitive
16	*pà-			location on
17	*kù-			location at
18	*mù-			location in
19	*pì-			diminutive

There are a few semantic regularities in Xhosa consistent with Table 1, among them that only [human] nouns bear the *um-/aba-* prefixes of classes 1/2, and kinship terms are consistently found in its subclasses 1a/2a bearing the prefixes *u-/oo-*. But these are one-way correlations: [human] nouns appear in all the classes, and 1a/2a include letters of the alphabet and an assortment of animals and inanimates. Other semantic associations reconstructed in Table 1 are weak or lost. Diminutive, augmentative, and locative classes 16 and 18 are absent in Xhosa; Nguni locative class 17 is prepositional (Buell 2007) and will not be relevant here. In what follows I overview the distribution of the major semantic fields across classes.

2.2.2 The heterogenous contents of Xhosa classes

(28) lists representative examples of the distribution of [human] nouns in Xhosa. In classes 3/4 there are very few and they are stigmatized,¹⁰ but in 1/2, 5/6, 7/8, and 9/10 there are many,

¹⁰ An anonymous reviewer asks if the stigma for [human] nouns in classes 3/4 might reduce to its proposed lack of semantic core. We will see that 3/4 and 5/6 pattern alike on my diagnostic for this lack, but 5/6 include a number of [human] nouns. I propose in §5.1 that in addition to interpretable gender ‘flavors,’ meta-linguistic mapping principles yield smaller strands of semantic coherence within genders; one such rule might assign stigmatized humans to class 3/4. Similarly, animals in classes 3/4 are few and appear to have in common distinctive projections from their heads, suggesting another such mapping: in addition to ‘rhino,’ ‘rooster,’ and ‘rabbit,’ ‘whale’ is in 3/4 perhaps due to its characteristic plume of spray.

without predictable semantic connotations positive or negative.¹¹

(28) Humans: classes 1/2

- | | | |
|--|--|--|
| a. um -fazi/ aba -fazi
1-woman/2-women
'woman/women' | b. um -ntu/ aba -ntu
1-person/2-people
'person/people' | c. um -/ aba -mntwana
1-child/2-children
'child/ren' |
|--|--|--|

Humans: classes 3/4 (few; stigmatized)

- | | | |
|---|---|--|
| d. um -gulukudu/ imi -gulukudu
3-gangster/4-gangster
'thug/s, gangster/s' | e. um -gewu/ imi -gewu
3-criminal/4-criminal
'criminal/s' | f. um -lwelwe/ imi -lwelwe
3-/4-disabled or poor
'disabled or poor person/s' |
|---|---|--|

Humans: classes 5/6

- | | | |
|---|--|---|
| g. i -qhawe/ ama -qhawe
5-hero/6-hero
'hero/es' | h. i -gqwetha/ ama -gqwetha
5-lawyer/6-lawyer
'lawyer/s' | i. i -sela/ ame -sela
5-thief/6-thief
'thief/thieves' |
|---|--|---|

Humans: classes 7/8

- | | | |
|---|---|--|
| j. isi -bonda/ izi -bonda
7-headman/8-headmen
'headman/men' | k. isi -hlobo/ izi -hlobo
7-relative/8-relatives
'relative/s' | l. isi -fundiswa / izi -fundiswa
7-scholar/8-scholar
'scholar/s' |
|---|---|--|

Humans: classes 9/10

- | | | |
|---|---|---|
| m. in -tombi/ iin -tombi
9-young.lady/10-young.lady
'young lady/ladies' | n. in -gcali/ iin -gcali
9-expert/10-expert
'expert/s or scientist/s' | o. im -bongi/ iim -bongi
9-poet/10-poet
'poet or praise singer/s' |
|---|---|---|

[Animal] nouns are also present in all pairs of classes as (29) demonstrates, though 3/4

include very few. The plurality are found in classes 9/10.¹²

(29) Animals: classes 1a/2a

- | | | |
|---|--|---|
| a. u -nokala/ oo -nokala
1a-crab/2a-crab
'crab/s' | b. u -krebe/ oo -krebe
1a-shark/2a-shark
'shark/s' | c. u -dyakalashe/ oo -dyakalashe
1a-jackal/2a-jackal
'jackal/s' |
|---|--|---|

Animals: classes 3/4 (few)

- | | | |
|--|---|--|
| d. um -khombe/ imi -khombe
3-rhino/4-rhino
'rhino/s' | e. um -qhagi/ imi -qhagi
3-rooster/4-roosters
'rooster/s' | f. um -vundla/ imi -vundla
3-rabbit/4-rabbits
'rabbit/s' |
|--|---|--|

¹¹ Mletshe (2019) illustrates some semantic regularities in the output classes of deverbal nouns which I take to be properties of the nominalizing suffix vowels involved, which are likely special choices of *n* (as proposed in Fuchs & Van der Wal 2022, Halpert & Hammerly 2025). Space precludes fuller discussion.

¹² A reviewer asks for a source. I have not found one for Xhosa though see Segerer (2025) on this generality in Bantu. Of 62 animals and insects for which I found Xhosa terms in Google Translate, 38 = 61% are in classes 9/10; the 2nd largest number 11/60 are in classes 5/6 (I exclude borrowings since their default class is 9/10).

Animals: classes 5/6

g. i-cikilishe/ama-cikilishe
5-lizard/6-lizard
'lizard/s'

h. i-hashe/ama-hashe
5-horse/6-horse
'horse/s'

i. i-hobe/ama-hobe
5-dove/6-dove
'dove/s'

Animals: classes 7/8 (few)

j. isi-gcawu/izi-gcawu
7-spider/8-spider
'spider/s'

k. isi-khova/izi-khova
7-owl/8-owl
'owl/s'

l. isi-lwanyana/izi-lwanyana
7-animal/8-animal
'animal/s'

Animals: classes 9/10 (plurality of animals; a fairly predictable mapping)

m. in-ja/izin-ja
9-dog/10-dog
'dog/s'

n. in-dlovu/iin-dlovu
9-elephant/10-elephant
'elephant/s'

o. i-hagu/ii-hagu
9-pig/10-pig
'pig/s'

[Inanimate] nouns are similarly scattered. For example, though many are found in 7/8, the generic term for 'thing(s)' is a class 9/10 noun.

(30) Inanimates: 1a/2a

a. u-lolilwe/oo-lolilwe
1a-train/2a-train
'train/s'

b. u-matshini/oo-matshini
1a-machine/2a-machine
'machine/s'

c. u-L/oo-L
1a-L/2a-L
'letter L/s'

Inanimates: 3/4

d. um-bhobho/imi-bhobho
3-pipe/4-pipe
'pipe/s'

e. um-nqwazi /imi-nqwazi
3-hat/4-hat
'hat/s'

f. um-ba/imi-ba
3-issue/4-issues
'issue/s'

Inanimates: 5/6

g. i-qhosha/ama-qhosha
5-button/6-button
'button/s'

h. i-cepe/ama-cepe
5-spoon/6-spoon
'spoon/s'

i. i-gama/ama-gama
5-word/6-word
'word/s or name/s'

Inanimates: 7/8

j. isi-bane/izi-bane
7-lamp/8-lamp
'lamp/s'

k. isi-tya/izi-tya
7-dish/8-dish
'dish/es'

l. isi-Xhosa
7-Xhosa
'Xhosa language'

Inanimates: 9/10

m. in-to/izin-to
9-thing/10-thing
'thing/s'

n. in-cwadi/iin-cwadi
9-book/10-books
'book/s'

o. in-daba/iin-daba
9-news/10-news
'news item/s'

The generic term for 'tree' is *um-thi*, a noun of classes 3/4 as is consistent with the proto-Bantu reconstruction in Table 1. The generic term for 'plant' is a deverbal noun of classes 7/8 (*isi-tyalo* < *uku-tyala* — 'to plant'), and generic 'bush' is a noun of 5/6. To the best of my knowledge there are no tree or plant terms in classes 1a/2a.

(31) Plants and trees: 3/4

- | | | |
|--|--|--|
| a. um -thi/ imi -thi
3-tree/4-trees
'tree/s' | b. um -hlonyane/ imi -hlonyane
3-Wormwood/4-Wormwood
'Wormwood tree/s' | c. um -nquma/ imi -nquma
3-wild.olive/4-wild.olive
'wild olive tree/s' |
|--|--|--|

Plants and trees: 5/6

- | | | |
|--|---|---|
| d. i -khala
5-aloe
'aloe/s' | e. i -tyholo/ ama -tyholo
5-bush/6-bushes
'bush/es' | f. i -cuba
5-tobacco
'tobacco' |
|--|---|---|

Plants and trees: 7/8

- | | | |
|--|---|--|
| f. isi -thungwa/ izi -thungwa
7-sew.PASS/8sew.PASS
'Cape Quince' (plaited for use) | g. isi -tyalo/ izi -tyalo
7-plant.NOM/8-plant.NOM
'plant/s' | h. isi -Qwane/ izi -Qwane
7-Protea/8-Protea
'Protea' |
|--|---|--|

Plants and trees: 9/10

- | | | |
|--|---|---|
| i. in -tshilo/ iin -tshilo
9-TSHILO /10-TSHILO
'Hedge Caper' | j. in -yibiba/ iin -yiba
9-lily/10-lillies
'lily/lillies' | k. in -gcongolo/ iin -gcongolo
9-reed/10-reeds
'reed/reeds' |
|--|---|---|

(32) shows that liquid and mass nouns are scattered across the Xhosa noun classes both singular and plural, apart from 1/2 and 1a/2a.

(32) Liquids and masses: scattered

- | | | | | | |
|---|--|--|---|-------------------------------------|--|
| a. um -chamo
3-urine
'urine' | b. i -gazi
5-blood
'blood' | c. ama -nzi
6-water
'water' | d. isi -dudu
7-porridge
'porridge' | e. i -oili
9-oil
'oil' | f. u -bisi
11-milk
'milk' |
| g. um -gubo
3-flour
'flour' | h. ama -zimba
6-sorghum
'sorghum' | i. i -tyuwa
9-salt
'salt' | j. u -daka
11-mud
'mud' | | |

Clauses and gerunds belong to a dedicated class 15 (see (33)), and pair with class 15 *ku*-agreement. This is not the default agreement for Xhosa conjoined nominals, unlike what has been reported for various other languages (see discussion in Adamson & Anagnostopoulou 2025, Corbett 1991, and Kramer 2015 on default gender agreement with clauses).¹³

¹³ Two related points: (i) Curiously, class 15 agreement *is* the default with Nguni low subjects in VS constructions (Carstens & Mletshe 2015, Halpert 2015, Pietrazko 2021), and in Zulu but not Xhosa, it serves as default with conjoined mismatched DPs (Bosch 1985). (ii) CP objects conjoin in the same way as nominal objects and their conjunction takes the same *ku*-object marker as a singleton CP (see (33)a). Conjoined CP subjects are judged very awkward, though conjoined gerundive subjects like (35)b are fine; agreement is *ku*- (there is no plural class for CPs and gerunds). Thanks to an anonymous reviewer for raising this question.

- (33) Clauses a. Ndi-(ya-**ku**)-cinga ukuba u-hambile u-Aviwe (nokuba u-fikile u-Mlu).
class 15 1sSM-DISJ-**15**om-think that 1sM-left 1a-Aviwe and.that 1sM-arrived 1a-Mlu
I think that Aviwe left (and that Mlu arrived).’
- Gerunds b. Uku-cula (noku-danisa) **ku**-mnandi.
class 15 15-sing (and.15-dance) **15**sM-nice
‘Singing (and dancing) is nice.’

An anonymous reviewer asks for a quantitative overview as evidence that all of noun classes 1-10 are active in the synchronic grammar, and none are instead small collections of fossilized forms. Table 2 gives the breakdown of nouns by class in two published studies of Xhosa corpora drawn from documents on the South African website including press releases, transcriptions of speeches, health information and discussion of government departments and services (results of the two studies are averaged). Classes 9/10 are the largest, consistent with the informal impression that they include many loan words, humans, animals, and miscellany. The spread of tokens across classes 1-8 shows that they are robust.^{14, 15}

Table 2: Occurrence of Xhosa noun classes in two corpora from the South African Government website (Gaustad & Puttkammer 2021; Gaustad & McKellar 2024)

1(a)/2(a)	3/4	5/6	7/8	9/10	11	14	15
12.1%	12.2%	16.45%	10.95%	35.3%	10.3%	2.4%	.25%

3 Agreement with conjoined singulars

3.1 “Default”/semantic agr with mismatched [sg+sg]: classes 2 and 8

Given their highly varied contents, Xhosa noun classes seem not to be characterized by consistent semantic features apart from a few limited strands of coherence; but as §1 pointed out, agreement with conjoined singulars reveals a conceptual division of [human] vs. others.

¹⁴ Gaustad & Puttkamer (2022) explain that the South African domain websites were crawled for documents available in all of Xhosa, Zulu, Swati and Ndebele with the goal of providing data usable for comparative work on these languages in NLP, HLT, and morphological analyses. “The retrieved documents in pdf and Microsoft Word format were converted to text format, language identification was performed ... after clean-up, the content was split into paragraphs, sentences, and tokens before the linguistic annotations were added...The final dataset contains 1431 paragraphs (for each of the languages) on different topics...”

¹⁵ Class 15 (infinitives and gerunds) is also robust though apparently scarce in this domain.

We see this manifested in the Xhosa (9)a,b and in (34)a,b where singular nouns from different classes are conjoined. In (34)a where the conjoined nouns denote inanimates, agreement is class 8. For conjoined [human] nouns of mismatching classes agreement is class 2.¹⁶

- (34) a. Um-nqathe ne-qanda **zi**-se tafile-ni. [Xhosa; Mitchely 2015:115 adapted]
 3-carrot and.5-egg 8SM-LOC table-LOC
 'The carrot and the egg are on the table.'
- b. Um-gulukudu ne-polisa **ba**-ya-baleka.
 3-gangster and.5-policeman 2SM-DISJ-run
 'The gangster and the policeman are running.'

Zi-agreement with mismatched inanimates like (34)a is actually ambiguous between classes 8 and 10, but in languages lacking this syncretism agreement with mismatched inanimates is clearly class 8. The pattern is common across Bantu languages (Corbett 1983, 1991, 2006, Corbett & Mtenje 1987, De Vos & Mitchley 2012, Diercks et al 2015, Givon 1970, Marten 2000, 2005, Roberts & Wolontis 1974, Schadeberg 2001, Simango 2012, and Voeltz 1971, Mitchley 2015). (35) and (36) provide examples from the literature (glosses adapted).

- (35) a. Mbale na ka-temo **v**-a-sowa. [ciNsenga; Simango 2012: 178]
 9.plate and 12-axe 8SM-past-miss
 'The plate and the axe are missing.'
- b. Li-doda na m-simbi **w**-e-fwika mailo [ciNsenga; Simango 2012:179]
 5-man and 1-girl 2SM-PST-arrive yesterday
 'The man and the girl arrived yesterday.'
- (36) a. U-mu-bíírá ní-í-ri-gena m-**bí**-síir:e [Kuria; from Diercks et al 2015:31]
 3-3-ball and-5-5-stone FOC-8SM-disappear.PST
 'The ball and the stone disappeared.'
- b. Axi-kelema no buchara **vo** vulavula. [Xitsonga; Mitchley 2015:65]
 DEF-7-scoundrel and 9.bucher 2SM.PRES talk
 'The scoundrel and the butcher are talking.'

Semantic underpinnings are clear: 1/2 are reconstructed as the canonical domain of [human] nouns, as Table 1 indicates, and classes 7/8 as that of mixed contents or [inanimate] nouns; §6 provides discussion and references.

¹⁶ The conjunction *na*-cliticizes to the following noun and there is vowel coalescence: a+i=e, a+u=o.

- c. Um-khonto nom-bhobho ✓*zi-* /X*i-*nomhlwa.
 3-spear and.3-pipe 8*SM*/ 4*SM*-have.3rust
 'The spear and the pipe are rusty.'
- d. Imi-nqwazi X*zi-* /✓*i-*se tafile-ni. [plural of N_{cl.3} is cl 4: *i-*agr]
 4-hats 8*SM*-/ 4*SM*-be table-LOC
 'The hats are on the table.'

Interviews of fifteen Xhosa speakers on the 3 examples of conjoined non-human-denoting nouns in (38)a-c yielded 33/45 choices of *zi-*, 4 of apparent single conjunct agreement *u-*, 5 cases where speakers offered both *zi-* and *u-* as equally good, one choice of gender-matching class 4 *i-*, and two of class 6 *a-*. Table 1 summarizes these results (in the interests of simplicity I indicate minority prefix choices only in tables, not in examples).

Table 3: 15 speakers x 1 [human], x 3 non-human

[3&3]	matching	default	single conjunct	either/or	other
human	0%	<i>ba-</i> 100%	0%	0%	0%
non-human	<i>i-</i> 2.2%	<i>zi-</i> 73.3%	<i>u-</i> 9%	<i>zi/u-</i> 11%	<i>a-</i> 4.4%

While Taraldsen et al give results identical to my own for [3&3] [human] in (37)a, they report significantly greater variation on the conjunction of non-human-denoting nouns in (39): 10/21 speakers chose *zi* in this case, 5/21 single conjunct agreement *u-*, 2 *zi/u-*, 1/21 a choice of *zi-* or class 6 *a-*, and 3/21 class 2 *ba-*. The latter is a significant difference from my own results where the canonically [human] *ba*-agreement did not occur once. Table 4 summarizes (% rounded).

- (39) Um-bingqo nom-nqathe ___se tafile-ni. [Taraldsen et al 2018: 1348]
 3-skirt and.3-carrot ___be table-LOC
 'A skirt and a carrot are on the table.'

Table 4: Taraldsen et al (2018)

[3&3]	matching	default	single conjunct	either/or	other
human	0%	<i>ba-</i> 100%	0%	0%	0%
non-human	0%	<i>zi-</i> 48%	<i>u-</i> 24%	<i>u-/zi-</i> 9.5%	<i>a/zi-</i> 4.5 %; <i>ba-</i> 14%

Lacking access to Taraldsen et al's speakers I can only offer a conjecture about the source of our divergent results. Several speakers I consulted were unfamiliar with the term *um-bingqo* which led me to replace the example in my interviews, and which suggests that this lexical item might

Table 5: My 15 speakers x 2 examples of each type

[5&5]	matching	default	single conjunct	either/or	other
human	0%	<i>ba-</i> 63.33%	<i>li-</i> 13.33%	<i>ba/li-</i> 20%	<i>zi-</i> 3.33%
non-human	0%	<i>zi-</i> 73.33%	<i>li-</i> 10%	<i>zi/li-</i> 13.33%	<i>zi/a-</i> 3.33%

Taraldsen et al's results for non-human [5&5] are much like my own, but for [human] [5&5] Taraldsen et al tested (42), with more mixed results than mine for (40)a,b: 8/20 *ba-*, 2 *ba/zi*, 1 *ba/li*, 5/20 *zi*, 3/20 SCA *li*, 1 *zi/ba* and 1 *zi/li* (see Table 6). *Zi-* is unexpected, since it is neither gender-matching nor the default for conjoined [human] nouns. As in the case of class 3 *um-bingqo*, it seems plausible that the lexical items were a confounding factor: class 9 nouns are also *i-*initial, so if *i-gosa* and/or *i-gqirha* were at all unfamiliar their phonological shapes might lead to class 10 plural agreement *zi-*. (§3.3.2 Table 12 will show that gender-matching plural agreement was the majority choice with [human] class 9 nouns in Taraldsen's study). I chose other examples because a couple of speakers spontaneously mentioned that class 1 *u-gqirha* is a more common word for 'doctor' than class 5 *i-gqirha* – '(divine/traditional) healer.'

- (42) I-gqirha ne-gosa ___sebenza ndawonye.
 5-healer and.5-steward work 9place.one
 'The healer and the steward are working together.'

Table 6 Taraldsen et al (2018)

[5&5]	matching	default	SCA	either/or	other
human	0%	<i>ba-</i> 35%	<i>li-</i> 10%	<i>ba/li-</i> 5%	<i>ba/zi-</i> 15%; <i>zi-</i> 30%; <i>li/zi-</i> 5%
non-human	0%	<i>zi-</i> 75%	<i>li-</i> 20%	0%	<i>zi/a-</i> 5%

Taraldsen et al (2018) conclude from the failures of gender-matching plural agreement with [3&3] and [5&5] that Bantu singular/plural pairings do not in fact share gender features; rather, each singular and each plural class is a distinct gender. However the Xhosa facts in (38)-(41) are, in reality, compatible with the gender analysis of pairs of classes owing to a similar pattern of gender-match failure in a language with a canonical three-gender system: in Bosnian, Croatian, and Serbian (BCS), neuter singulars do not pair with neuter plural agreement but rather with

default masculine.

- (43) Sel-o i polj-e su poltopljen-i /*poltopljen-a. [BCS]
 village-NEUT.SG and field.NEUT.SG AUX.PL flooded-✓MASC.PL/ *NEUT-PL
 'The village and the field are flooded.'

3.3 Where regular plural agreement obtains

3.3.1 Conjoined singulars [1&1], [7&7], [9&9] may pair with matching plural agreement or default

As mentioned in §1, agreement does match, in certain genders. 100% of all speakers consulted accepted only class 2 *ba*-agreement with the uniform [human] class 1 conjuncts like (44). Since *ba*- is both matching and default for [human], I merge the two categories in Tables 7 and 8.

- (44) Um-ntwana nom-fazi **ba**-ya-dlala. [1&1=2]
 1-child and.1-woman 2SM-DISJ-play
 'The child and the woman are playing.'

Turning to non-humans, Taraldsen et al report that in (45)a, 14/20 speakers chose only *ba*-, 3 proposed *ba*- and singular *u*- as comparably acceptable, 1 speaker proposed both *ba*- and default *zi*-, 1 speaker suggested only class 8 *zi*-, and another class 6 *a*-. My results: in 22/30 cases speakers offered *ba*- in (45)a and (45)b; the remainder offered class 8 *zi*- apart from one single conjunct *u*- ((45)a from Taraldsen et al; (45)b added in my own research).

- (45) a. U-L no-M **ba**-se tafile-ni. [1&1=2]
 1a-L and.1a-M 2SM-LOC table-LOC
 'The L and the M are on the table.'
- b. U-loliwe kunye no-matshini **ba**-ya-hamba. [1&1=2]
 1a-train and and.1a-machine 2SM-DISJ-move
 'The train and the machine are moving.'

Table 7: 15 speakers x 2 examples in each category

[1&1]	matching	default	single conjunct	either/or	other
human	<i>ba</i> - 100%	0%	0%	0%	0%
non-human	<i>ba</i> - 73%	<i>zi</i> - 23.33%	<i>u</i> - 3.33%	0%	0%

Table 8 Taraldsen et al: 20 speakers

[1&1]	matching	default	single conjunct	SCA/default	match/default; other
human	<i>ba</i> - 100%	0%	0%	0%	0%
non-human	<i>ba</i> - 70%	<i>zi</i> - 5%	0%	<i>u/ba</i> - 15%	<i>ba/zi</i> 5%; cl 6 <i>a</i> - 5%

Similarly, in the majority of cases [7&7] pairs with *zi*-agreement. For non-human [7&7] such as (46), speakers offered only *zi* (I again merge matching and default categories since *zi*- could be either for non-human [7&7]). For the conjoined [human] nominals in (47)a, Taraldsen et al report that 20/21 speakers chose *zi*-; a single consultant offered both *zi*-/*ba*-. My results are less uniform for the same and additional examples of [human] [7&7]: on (47)a,b there were 18/30 choices of *zi*-, 3/30 either *ba*- or *zi*-, 9/30 choices of class 2 *ba* alone ((46)a and (47)a from Taraldsen et al 2018:1345). Tables 9 and 10 summarize.

- (46) a. Isi-bane nesi-tya **zi**-nyamalele. [7&7=8]
 7-lamp and.7-dish 8SM-disappeared
 'The lamp and the dish have disappeared.'
- b. Isi-Xhosa nesi-Zulu **zi**-theth-wa e-Mzantsi Afrika.
 7-Xhosa and.7-Zulu 8SM-speak-PASS LOC-South Africa
 'Xhosa and Zulu are spoken in South Africa.'
- (47) a. Is-anuse nes-angoma **zi**-sebenza ndawonye.
 7-diviner and.7-healer 8SM-work 9place.one
 'The diviner and the healer work together.'
- b. Isi-bhanxa nes-azi **zi**-ya-funda
 7-fool and.7-scholar 8SM-DISJ-study
 'The fool and the scholar are studying.'

Table 9 Agreement with [7&7], /15 speakers x 2 examples

[7&7]	matching	default	either/or	single conjunct	other
human	<i>zi</i> - 60%	<i>ba</i> - 30%	<i>zi/ba</i> - 10%	0%	0%
non-human	<i>zi</i> - 100%		0%	0%	0%

Table 10 Taraldsen et al: /20 speakers

[7&7]	matching	default	either/or	single conjunct	other
human	<i>zi</i> - 95%	<i>ba</i> - 0%	<i>zi/ba</i> - 5%	0%	0%
non-human	<i>zi</i> - 100%		0%	0%	0%

Turning to [human] [9&9], in my interviews agreement choice on this combination was the least predictable. As summarized in Table 9, *zi*- was the sole choice in 15/30 instances and one of two options *zi/ba* in 3/30 more; there were 12/30 choices of *ba*-. For conjoined class 9 non-humans like (49)a,b, *zi*- was the only agreement offered. Given that this is the expected form

for both matching plural class 10 and default non-human class 8, and in light of the split between matching *zi-* and human default *ba-* for human-denoting nouns, it seems likely that *zi-* agreement with [9&9] non-human varies between matching class 10 and default class 8.¹⁹

- (48) a. In-tombi nem-bongi **zi-/ba-**ya-cula. [9&9=10 or 2]
 9-girl and.9-poet 10/2SM-DISJ-sing
 'The girl and the poet are singing.'
 b. In-gcaphephe nen-gcali a-**zi-/ba-**vumel-an-i.
 9-expert and.9-specialist NEG-10/2SM-agree-RECIP-NEG
 'The expert and the specialist disagree.'
- (49) a. In-cwadi nepeni **zi-se** tafile-ni. [9&9=10]
 9-book and.9-pen 10SM/8SM-LOC table-LOC
 'The book and the pen are on the table.'
 b. In-dlovu (kunye) nen-gwe **zi-ya-lwa**.
 9-elephant and and.9-leopard 10SM-DISJ-fight.
 'The elephant and the leopard are fighting.'

Table 11: 15 speakers x 2 examples each type

[9&9]	matching	default	match/default	single conjunct	match/SCA
human	<i>zi-</i> 50%	<i>ba-</i> 40%	<i>zi/ba-</i> 10%	0%	0%
non-human	<i>zi-</i> 100%		0%	0%	0%

Table 12 shows that Taraldsen et al's findings were significantly more uniform: they report 16/20 tokens (80%) of *zi-agr* in (48)a as either the sole choice or one of two options, the other being single conjunct agreement or default.²⁰

¹⁹ An anonymous reviewer suggests treating *zi*-agreement as consistently class 8 and never 10, but there is no motivation for assuming that [human] [9&9] would pair with default class 8 rather than either gender-matching class 10 or [human] default class 2. The reviewer suggests as an alternative that *zi*-agreement with [human] [9&9] is formal, grammatical class 10 agreement, while *zi*-agreement with conjoined non-humans is always default class 8. But if formal agreement is possible for [9&9] humans, I can see no basis for it to systematically fail with non-humans, triggering default 8-AGR across the board. A plausible implementation of the 10=8 idea might be that class 10 has merged with class 8. Assuming genders underlie paired classes, this entails that gender E no longer exists in Xhosa; instead, gender D has two sets of singular nominal prefixes (gender D1: *isi /izi*; gender D2: *i(n) / ii(n)* < 9/10) and two sets of singular agreement markers (D1: *si-*, D2: *i-* < class 9). This is conceivable but sheds no light on the question of why uniform conjoined singulars of some genders pair with matching agreement and others do not; in addition, the rates of matching agreement differ; matching is more common with [7&7], supporting a gender difference. Absent analytical advantages to this hypothesis I continue to assume that 7/8 and 9/10 are distinct genders with syncretic plural agreement.

²⁰ In (48)a [human] [9+9], Taraldsen et al report that 12/20 speakers offered only *zi-*, 3 proposed both *zi* and SCA *i-*, 1 more offered both *ba* and *zi*, yielding a total of 16/20 offering *zi*; 4/20 speakers offered only *ba-*.

Table 12 Taraldsen et al: 20 speakers

[9&9]	matching	default	single conjunct	match/default	match/SCA
human	zi- 60%	ba- 20%	0%	zi-/ba- 5%	zi or i- 15%
non-human	zi- 100%		0%	0%	0%

3.4 Summary and discussion

Tables 13 and 14 present the full sets of results.

Table 13 Combined results: my speakers

Class	matching	default	SCA	either/or	other
[3&3] human	0%	100%	0%	0%	0%
[3&3] non-human	2.2%	73.3%	9%	DEF/SCA 11%	cl 6 a- 4.4%
[5&5] human	0%	63.33%	13.33%	DEF/SCA 20%	zi- 3.33%
[5&5] non-human	0%	73.33%	10.33%	DEF/SCA 13.33%	zi/a- 3.33%
[1&1] human	100%		0%	0%	0%
[1&1] non-human	73%	23.33%	3.33%	0%	0%
[7&7] human	60%	30%	0%	10%	0%
[7&7] non-human	100%		0%	0%	0%
[9&9] human	50%	40%	0%	MATCH/DEF 10% (zi/ba)	0%
[9&9] non-human	100%		0%	0%	0%

Table 14 Combined results: Taraldsen et al (2018)

Class	matching	default	SCA	either/or	other
[3&3] human	0%	100%	0%	0%	0%
[3&3] non-human	0%	48%	24%	DEF/SCA 9.5%	a/zi- 4.5%; ba- 14%
[5&5] human	0%	35%	10%	DEF/SCA 5%	ba/zi- 15%; zi- 30%; li/zi- 5%
[5&5] non-human	0%	75%	20%	0%	zi/a- 5%
[1&1] human	100%		0%	0%	0%
[1&1] non-human	70%	5%	0%	DEF/SCA 15% (zi/u) MATCH/DEF 5% (ba/zi)	a- 5%
[7&7] human	95%	0%	0%	MATCH/DEF 5%	0%
[7&7] non-human	100%		0%	0%	0%
[9&9] human	60%	20%	0%	MATCH/DEF 5% (zi/ba) MATCH/SCA 15% (zi/i)	0%
[9&9] non-human	100%		0%	0%	0%

I set aside Taraldsen et al's very mixed results for non-human [3&3] and human [5&5], which I suggested in §3.2 and §3.3 reflect confounds of the lexical items in his test sentences.

I also set aside choices offered $\leq 5\%$ of cases; I will not attempt to explain these though it is noteworthy that in several cases they involved class 6, which has been reconstructed as a polyplural, working for more than one singular class.²¹

Lastly, I set aside apparent SCA as due to an ambiguity of Xhosa *na* between ‘and’ and ‘with.’ This is consistent with the fact that SCA is incompatible with collective predicates, which would not be expected were the structure for it the same as for resolved agreement; speakers who accept (50)a nonetheless reject SCA in (50)b,c (same for singular agreement on ‘collide’ with [3&3] and [5&5] inanimates, omitted for reasons of space). I suggest that (50)a has the structure in (51)b and loosely paraphrases (51)a.²²

- (50) a. I-gqwetha ne-sela li-phum-ile. SCA
 5-lawyer and.5-thief 5SM -leave-PST.DISJ
 ‘The lawyer and the thief left’
- b. I-gqwetha ne-sela a- ba/*li/*a-vumel-an-i. *SCA
 5-lawyer and. 5-thief NEG-2SM/*5SM/6SM-agree-RECIP-NEG
 ‘The lawyer and the thief disagree with each other.’
- c. I-gqwetha ne-sela ba/ *l/ *a-a-zi-bon-a e-sipili-ni. *SCA
 5-lawyer and.5-thief 2SM /*5SM/6SM- PST-SELF-see LOC-mirror-LOC
 ‘The lawyer and the thief saw themselves in the mirror.’
- (51) a. I-gwetha li-phum-ile ne-sela.
 5-lawyer 5SM -leave-PST.DISJ. with.5-thief
 ‘The lawyer left with the thief.’
- b.
-
- ```

graph TD
 DP1_1[DP1] --- DP1_2[DP1]
 DP1_1 --- PP[PP]
 DP1_2 --- lawyer[lawyer]
 PP --- P[P]
 PP --- DP2[DP2]
 P --- with[with]
 DP2 --- thief[thief]

```

Those exclusions leave the Xhosa patterns in (52) to be captured.

<sup>21</sup> See Maho (2003). This pattern has echoes in modern Xhosa where groups of people are pluralized with class 6 regardless of their singulars: *in-/ama-doda* – ‘9man/6men’; *um-/ama-Zulu* – ‘1Zulu person/6people’.

<sup>22</sup> There are several puzzles associated with *na*: a single speaker consistently judged that in [X&Y] with SCA, X and Y act separately, which suggests clausal coordination and ellipsis (Aoun, Benmamoun & Sportiche 1994, van Alem 2023) but other speakers disagreed; Taraldsen et al included *ndawonye* to avoid such readings, glossed as ‘together’ but in fact it means ‘in the same place’ and for the speaker who reports the readings it is compatible with them. DP-internally *na* is somewhat degraded as ‘with’ so rejected by many; under the PP analysis of *na*, why ‘the lawyer’ can’t antecede the generic reflexive in (50c) is unclear to me. A last puzzle, ‘The man sees each other *na* the woman’ is a fine paraphrase for ‘The man and the woman see each other’ in Xhosa. Further work on *na*-constructions is needed but lies outside the scope.

- (52) a. [3&3], [5&5] generally cannot pair with gender-matching plural agreement.  
 b. In contrast, with [1&1], [7&7], [9&9] gender-matching plural agreement is the majority choice.  
 c. Class 2 *ba-* and class 8 *zi-* provide default agreement for [human] and [inanimate] conjunctions respectively whether [3&3], [5&5], or non-uniform [ $X_{sg}$  &  $Y_{sg}$ ].  
 d. Where matching agreement is possible, default agreement is also an option.

It seems an unlikely coincidence that two genders exhibiting gender-matching plural agreement with conjoined, gender-matched singulars are also the sources of default agreements, as (53) asserts. Building on this I propose the broad generalization in (54).

- (53) That the same Xhosa plural classes serve as defaults and as gender-matching agreement with conjoined singulars is not accidental.  
 (54) Resolved agreement semantic conjecture: gender-matching plural agreement with conjoined singulars correlates non-coincidentally with the association of interpretable content to the gender.

Since [9&9] also licitly pair with matching plural agreement, a generalization across all three of the relevant classes is possible on the assumption that 9/10 also has a semantic core. I suggest that this is [animal] (equivalently, non-human animate) in light of animal terms' preponderance in 9/10 synchronically and in reconstructions (see conjoined animals with class 10 agreement in (49)b; note 12; discussion in §6 of Denny & Creider 1986; Aikhenveld 2019 on the large range of semantic parameters in noun categorization systems). Thus 1/2, 7/8, and 9/10 have the content [human], [inanimate], and [animal] respectively.

Building on (54) I propose that on a language-particular basis, a small number of genders with recognized association to salient categories of entities such as [human], [animal], [inanimate], [male], [female], have formal morphosyntactic legibility in agreement with conjuncts. I classify the former type of gender as having interpretable content, and the others as uninterpretable. It seems clear that what is or is not adopted as interpretable varies across languages somewhat arbitrarily. In Bantu languages, [male] and [female] are not grammaticized

in this way, and nor is [human] vs. [non-human] the core content of genders in well-studied Indo-European languages. Yet there is no evidence that these properties of entities are less salient than in languages where they are associated with specific genders.

Under this approach, genders in which matching plural agreement succeeds with conjoined singulars have something special going for them that other genders lack. In contrast, a common view is that matching agreement is the norm; a gender in which it systematically fails is taken to be exceptional. This perspective informs approaches to the contrast between neuter and other BCS genders. Recall that BCS neuter patterns with Xhosa [5&5] and [3&3] in disallowing matching plural agreement, in contrast with BCS feminine and masculine (see (43) repeated below vs. (55) and (56), from Murphy & Puskar 2018; glosses adapted). Despić (2016), Nevins (2018), Tsimpli & Hulk (2013) propose that gender-matching agreement fails for conjoined neuters because neuter is the absence of gender or negative values [-FEM, -MASC]; in the calculus of resolved agreement, adding two such values yields the empty set, triggering default.

(43) Sel-o i polj-e su poltopljen-i /\*poltopljen-a. [BCS]  
 village-NEUT.SG and field.NEUT.SG AUX.PL flooded-✓MASC.PL/\*NEUT-PL  
 'The village and the field are flooded.'

(55) Ovaj covek i onaj decak su stigli-i. [BCS: Despić 2017: 291]  
 this man.MASC and that boy.MASC are arrived.MASC.PL  
 'This man and that boy arrived.'  
 [&P MASC &MASC = MASC.PL]

(56) [&P Zavesa i biljka] su ukrašavale prozor.  
 curtain.F.SG and plant.F.SG are decorate.PRT.F.PL window  
 'A curtain and a plant decorate the window.'  
 [&P FEM &FEM = FEM.PL]

This seems a workable approach to a single outlier gender, but it does not extend in an obvious way to multiple outlier genders within a single language, as is the case in Xhosa.

§6 will review and reject for Xhosa a different approach to BCS that has more initial plausibility, proposed in Arsenijević (2017, 2021) and adopted in Adamson & Anagnostopoulou (2024): that a specific type of semantic content can make a gender incompatible with resolved

agreement. We will see in §6 that Xhosa classes 3/4 and 5/6, in which matching agreement fails, show no signs of defectiveness paralleling those Arsenijević identifies for BCS neuter.

Equally importantly, in the Shona language the genders in which conjoined singulars pair with default/semantic agreement outnumber by 6:2 those in which gender-matching agreement succeeds. The facts strongly support the proposed refocus to treat genders which *do* allow or require matching as the exceptional ones. The large number of genders in Shona affords this alternative insight. I turn to the Shona facts next.

### 3.5 Evidence from Shona

Recall from §2 that Shona includes eight singular/plural pairs of noun classes – three more than Xhosa -- plus a few singletons. All Shona nouns of classes 1/2 refer to humans.

Conjunctions [1&1] pair with matching class 2 agreement in Shona, just as in Xhosa:<sup>23</sup>

- (57) Mu-rume ne-mu-kadzi **va**-ri panze. [Shona: 1&1=2]  
 1-man and-1-woman 2SM-be outside  
 ‘The man and the woman are outside.’

[Human] and [inanimate] nouns of [3&3] and [5&5] pair only with semantic agreement of classes 2 and 8 respectively. As is true in Xhosa they cannot pair with matching agreement:

- (58) Mu-nwe ne-mu-romo Xi /√**zvi**-ne utachiona. [Shona: 3&3=8]  
 3-mouth and-3-finger 4SM/ 8SM-have 14infection  
 ‘The mouth and the finger are infected.’

- (59) Dombo ne-zai Xa /√**zvi**-ri panze. [Shona: 5&5=8]  
 5stone and-5egg 6SM/ 8SM-be outside  
 ‘The stone and the egg are outside.’

- (60) Benzi ne-dinga Xa- /√**va**-ri ku-shayikwa. [Shona: 5&5=2]  
 5fool and-5dimwit 6SM/ 2SM-be 15-missing  
 ‘The fool and the dimwit are missing.’

Conjoined [7&7] nouns pair with class 8 agreement if referring to non-humans. There are not a large number of [human] nouns outside of classes 1/2 in Shona; those of classes 7/8 are

<sup>23</sup> This section reports the consistent judgments of three Zimbabwean Shona speakers.

stigmatized or diminutive. When conjoined they pair readily with class 2 agreement, though class 8 is judged an acceptable if slightly less felicitous alternative.<sup>24</sup>

- (61) Chi-ngwa ne-chi-bage **zvi**-ri pa-tafura. [Shona: 7&7=8]  
 7-bread and-7-maize 8SM-be LOC-table [inanimate]  
 'The bread and the maize are on the table.'
- (62) Chi-dhakwa ne-chi-komana **va**-ri/?**zvi**- ku-netsana. [Shona: 7&7=2 or 8]  
 7-drunkard and-7-small.boy 2SM/ 8SM-be 15-argue [human]  
 'The drunkard and the small boy argued.'

Like [3&3] and [5&5], Shona [9&9] [human] nouns always take class 2 *va*-agreement as shown in (63)b, and non-humans take class 8 *zv*-agreement ((63)c-f), unlike plurals of class 9 nouns (see (63)a). This is a striking contrast to Xhosa wherein the majority of speakers pair [9&9] [human] with *zi*-, motivating my proposal that Xhosa *zi*- can be gender-matching plural agreement.

- (63) a. Im-bwa/n-herera **dz**-aka-on-ana. [Shona: 10agr = *dzi*-]  
 10-dogs/10-orphans 10SM-PST-see-RECIP  
 'The dogs/orphans saw each other'
- b. N-herera ne-n-yanzvi **va**- /\***dz**-aka-on-ana. [Shona: 9&9=2]  
 9-orphan and-9-expert 2SM-/\*10SM-PST-see-RECIP  
 'The orphan and the expert saw each other.'
- c. Im-bwa nhema ne-m-hou chena **zv**- /\***dz**-aka-rwa. [Shona: 9&9=8]  
 9-dog 9-black and-9-cow 9-white 8SM-/\*10SM-PST-fall  
 'The black dog and the white cow fell.'
- d. M-huno iyi ne nzeve iyi **zvi**- /\***dzi**-ne utachiona. [Shona: 9&9=8]  
 9-nose 9this and 9ear 9this 8SM-/\*10SM-have 14-infection  
 'This nose and this ear are infected.'
- e. M-biya iyi ne-sando iyo **zvi**- /\***dzi**-ne n-gura. [Shona: 9&9=8]  
 9-bowl 9-this and-9-hammer 9-that 8SM-/\*10SM-have 9-rust  
 'This bowl and that hammer are rusty.'
- f. N-yota ne-n-zara **zv**- /\***dz**-aka-tambudza va-fambi. [Shona: 9&9=8]  
 9-hunger and-9-thirst 8SM-/\*10SM-afflict 2-travelers  
 'Hunger and thirst afflicted the travelers.'

As these examples show, the contents of Shona classes 9/10 are not distinguished by any special semantic themes; they include the plurality of animals, some body parts, borrowings, a few exceptional humans, and abstracts like 'hunger' and 'thirst.' 9/10 seem the largest and

<sup>24</sup> Speakers volunteered class 2 agreement; upon questioning judged 8 acceptable though not as good as 2.

most diverse classes, just as in Xhosa.<sup>25</sup>

As noted in §2, Shona class 11 nouns pluralize in class 10. Like [9&9], [11&11] conjuncts combine with default class 8 *zvi-* and not with the class 10 *dzi-* which agrees with their plurals:

- (64) a. Ru-kova urwu no-ru-kova urwo **zvi-** /\***dzi-**asvibiswa. [Shona: 11&11=8]  
11-stream 11this and-11-stream 11that 8SM- /\*10SM-PERF.polluted  
'This stream and that stream are polluted.'  
b. Hova **dz-**akasvibiswa.  
10-stream 10SM-PERF.polluted  
'The streams are polluted.'

Nouns of class 14 pluralize in class 6, but conjoined class 14 singulars pair with class 8 *zvi-*, not with the class 6 *a-* subject marker that goes with their plurals (see (65)a,b):

- (65) a. U-ta ne-u-tanho **zvi-**ri/\***a-**ri kunze. [Shona: 14&14=8]  
14-bow and-14-ladder 8SM-/\*6SM-be outside  
'The bow and the ladder are outside.'  
b. Ma-uta ne-ma-utanho **a-**ri kunze.  
6-bows and-6-ladders 6SM-be outside  
'The bows and the ladders are outside.'

Diminutives are productively formed with the prefixes *ka-/tu-* of classes 12/13.<sup>26</sup> Their plurals pair with *tu-* agreement (see (66)a,b). But (67) shows that when diminutives conjoin, agreement must be taken from class 2 or 8.<sup>27</sup>

- (66) a. mu-sikana → ka-sikana b. Tu-sikana **tw-**a-nyangadika.  
1-girl 12-girl 13-girls 13SM-ASP-disappear  
'girl' 'tiny girl' 'The tiny girls disappeared.'  
(67) a. Ka-sikana ne-ka-kómáná **v-** /\***tw-**a-nyangadika. [Shona: 12&12=2]  
12-girl &-12-boy 2SM/\*12-ASP-disappear  
'The tiny girl and the tiny boy disappeared.'  
b. Ka-mba ne-ka-motokari **zv-** /\***tw-**a-nyangadika. [Shona: 12&12=8]  
12-house &-12-car 8SM/\*13SM-ASP-disappear  
'The tiny house and the tiny car disappeared.'

<sup>25</sup> No existing Shona corpus codes noun classes as in Xhosa Table 2. AI searches by noun class yield tokens in quantities from larger>smaller in 9/10>5/6>7/8, same as for Xhosa. 11/10 and 14/6 yield the fewest tokens.

<sup>26</sup> Dechaine et al (2014) argue that Shona class prefixes are multi-functional, having a range of derivational uses in addition to number-marking sg/pl. This state of affairs is compatible with the view that affixes can have both genders and meanings, while genders directly imbued with semantic content are few; see note 11.

<sup>27</sup> There is also augmentative formation correlating with classes 21/6, for which I presently lack data.

Summing up, in Shona the matching agreement pattern is the exceptional one. Taking [11/10] and [14/6] to be genders like other singular/plural pairings, Shona has nearly three times the number of genders as better-studied European languages and several more than the five main singular/plural pairings found in Xhosa. But the only consistent agreement patterns with conjoined singulars are class 2 [human] and class 8 [non-human]. The next section presents the tools and core ingredients for an analysis reflecting this state of affairs.

## 4 Gender and *n*P-stacking

### 4.1 Kramer's system

I turn to fleshing out the syntax of gender – an important building block in the analysis of agreement with conjoined singulars. As noted in §1, Kramer proposes that gender is a feature of the categorizer *n* (see (1)-(2), repeated below).

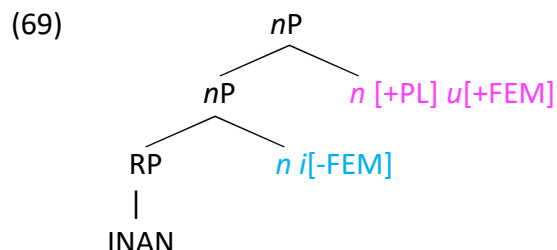
- (1) a. la       mujer               b. el        hombre       c. la       pluma       [Spanish]  
       the.FEM woman           the.MASC man       the.FEM pen  
       'the man'               'the woman'       'the pen'
- (2) a. [<sub>n</sub>P [<sub>VP</sub> MUJER] *n*<sub>[ifem]]</sub>   b. [<sub>n</sub>P [<sub>VP</sub> HOMBRE] *n*<sub>[i[-fem]]</sub>]   c. [<sub>n</sub>P [<sub>VP</sub> PLUMA] *n*<sub>[ufem]]</sub>]

I adopt this approach and a second tool from Kramer: gender stacking. Kramer follows Lecarme (2002) in assuming that pluralization imposes a gender on Somali nouns because in Somali, number is a feature of *n*. In (68), *inan* 'son' or 'boy' shifts from masculine to feminine when an *n* with the features *i*[+PL], *u*[+FEM] stacks above the *i*[-FEM] *n* of *inan* to create a plural that is grammatically feminine though semantically masculine as shown in (69). There is also a plural *n* with *u*[-FEM]; the *n* approach captures the fact that Somali number can impose a gender change and encodes which of eight pluralization strategies applies to a given noun.<sup>28</sup>

<sup>28</sup> Note that the interpretable gender is closest to the root in Kramer's account of Somali stacking, which will be true of my approach to Bantu. This is consistent with Puskar's (2017) approach to BCS hybrid agreement, but contra that of Landau's (2016) to Hebrew hybrid agreement among other works with hybrid as focus. Thanks to an anonymous reviewer for pointing this out; space precludes discussion of the alternatives.

- (68) a. ínan 'son, boy (m.)'  
b. inammó 'sons, boys (f.)'

[Somali]



## 4.2 ns for Xhosa

Recall the Bantu gender system proposed in Carstens (1991), reproduced below.

- (17) Bantu Genders (Carstens 1991)

Gender A: stems of Classes 1/2

Gender B: stems of Classes 3/4

Gender C: stems of Classes 5/6

Gender D: stems of Classes 7/8

Gender E: stems of Classes 9/10

Etc.

(70) provides an inventory of *ns* for Xhosa in which 1/2 and 7/8 have flavors *i*[human] and *i*[inanimate] respectively, and 9/10 has an *i*[animal] flavor reflecting its synchronic and reconstructed contents. All of these semantic properties entail [entity]; departing from the approach to arbitrary members in (2)c I propose that *i*[entity] is a building block of any *igender*, and that plain *i*[entity] flavors of each gender combine with its arbitrary members.<sup>29</sup> There is accordingly formal agreement available with uniform conjuncts [1&1], [7&7], and [9&9] whether arbitrary or not. In contrast genders B and C, underlying 3/4 and 5/6 respectively, lack any interpretable flavors, so they pair only with semantic agreement from 2, 8, or 10.

- (70) ns for Xhosa

Classes 1/2 = Gender A

*n*<sub>A1</sub> *i*[entity] *i*[human]

*n*<sub>A2</sub> *i*[entity] (for arbitrary members)

Classes 3/4 = Gender B

*n*<sub>B</sub> uninterpretable for all members

Classes 5/6 = Gender C

*n*<sub>C</sub> uninterpretable, for all members

Classes 7/8 = Gender D

*n*<sub>D1</sub> *i*[entity] *i*[inanimate]

*n*<sub>D2</sub> *i*[entity] (for arbitrary members)

Classes 9/10 = Gender E

*n*<sub>E1</sub> *i*[entity] *i*[animal]

*n*<sub>E2</sub> *i*[entity] (for arbitrary members)

<sup>29</sup> Ferrari (2005), Kihm (2005), Picallo (2008) analyze gender as always interpretable but along different lines.

A successful analysis must capture the semantic unpredictability of class memberships on the one hand and the predictable classes of default agreement with mismatched conjuncts and ‘outlier’ class [3&3] and [5&5] conjuncts on the other. To this end I propose that Xhosa *ns* are able to combine unusually freely not only with roots but with *nPs*, so that *nPs* of various genders routinely stack above *igender* cores, dispersing nouns from the various semantic fields across classes. As for borrowed or innovated nouns, I propose that these are formed in the same way; thus when new [human] nouns are coined or borrowed their roots combine at the lowest level with the *n<sub>A</sub>* which encodes the [human] semantic feature, but the resulting *nP<sub>A</sub>* is in many cases selected as complement by an *n* of a different gender.<sup>30</sup>

(71) Sample structures of [human] nouns: a core of 1/2 = gender A

- |    |                                                           |                                                              |                               |
|----|-----------------------------------------------------------|--------------------------------------------------------------|-------------------------------|
| a. | um-ntwana/aba-ntwana<br>1-child/2-children<br>‘child/ren’ | [ <i>n</i> <sub>1/2</sub> VMNTWANA]                          | in 1/2 for [human]            |
| b. | um-gewu/imi-gewu<br>3-criminal/4-criminal<br>‘criminal/s’ | [ <i>n</i> <sub>3/4</sub> [ <i>n</i> <sub>1/2</sub> VGEWU]]  | un of 3/4 stacks above in 1/2 |
| c. | i-butho/ama-butho<br>5-warrior/6-warrior<br>‘warrior/s’   | [ <i>n</i> <sub>5/6</sub> [ <i>n</i> <sub>1/2</sub> VBUTHO]] | un of 5/6 stacks above in 1/2 |

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<sup>30</sup> A reviewer asks if concord is possible in the class of the inner *n*. As far as I know Bantu exhibits this only in locatives like the Chichewa *pa chiseko chirichonse* ‘on every door,’ where *chirichonse* – ‘7every’ agrees with *chiseko* – ‘7door’ and 16pa – ‘on’ is an independent word (Carstens 1997). Carstens analyzes Chichewa locatives in terms of stacked DPs in which gender-agreeing Case-markers like *pa* introduce DP2; Fuchs & Van de Wal (2018) propose *pa* is an *n*, introducing a NumP. Both are levels of structural complexity absent in my proposed *nP* stackings. The reviewer asks also if there is overt prefix stacking to support this analysis as in Shona diminutives, where intrinsic class prefixes optionally appear below the diminutive prefixes (see (i)a from Dechaine et al 2014:41 and my §5.2 (99)); Fuchs & Van der Wal (2018) analyze this as optional spell-out of an inner *n*. I have analyzed class prefixes as number morphology, so prefix stacking is not predicted to accompany the *nP* stackings proposed here. Madzingwa & Kadenge (2014) report that stacking is obligatory when evaluative morphology is added to Shona deverbal nouns as shown in (i)b-d; three speakers I consulted confirmed this (though rejecting their additional claim that stacking is otherwise ruled out). The pattern suggests that an inner prefix may signal greater structural complexity of [DIMIN]’s complement, unavoidable in the case of derived nominals but optional with simple nouns; I leave details to future research.

|     |                                              |                                                         |                                                      |                                                         |         |
|-----|----------------------------------------------|---------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|---------|
| (i) | a. ka(mu)sikana<br>12-1-girl<br>‘small girl’ | b. ka-*(mu)bat-w-a<br>12-1-hold.PASS<br>‘small captive’ | c. ka-*(mu)-bat-i<br>12-1-hold-NOM<br>‘small captor’ | d. zi-*(mu)-cher-i<br>21-1-drink-NOM<br>‘heavy drinker’ | [Shona] |
|-----|----------------------------------------------|---------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|---------|

- d. isi-hlobo/izi-hlobo                    [*n*<sub>7/8</sub> [*n*<sub>1/2</sub> V<sub>HLOBO</sub>]]    in [entity] of 7/8 stacks above in 1/2  
7-relative/8-relative  
'relative/s'
- e. in-tombi/iin-tombi                    [*n*<sub>9/10</sub> [*n*<sub>1/2</sub> V<sub>TOMBI</sub>]]    in [entity] of 9/10 stacks above in 1/2  
9-young.lady/10-young.lady  
'young lady/ladies'

Similarly, [animal] nouns have gender E cores as in (72).

(72) Structure of [animal] nouns: an inner layer of 9/10 = gender E

- a. in-dlovu/iin-dlovu                    [*n*<sub>9/10</sub> V<sub>DLOVU</sub>]                    in 9/10 for [animal]  
9-elephant/10-elephant  
'elephant/s'
- b. u-nokala/oo-nokala                    [*n*<sub>1/2</sub> [*n*<sub>9/10</sub> V<sub>NOKALA</sub>]]    in [entity] of 1/2 above in of 9/10  
1a-crab/2a-crab  
'crab/s'
- c. um-vundla/imi-vundla                    [*n*<sub>3/4</sub> [*n*<sub>9/10</sub> V<sub>VUNDLA</sub>]]    un of 3/4 above in of 9/10  
3-rabbit/4-rabbit  
'rabbit/s'

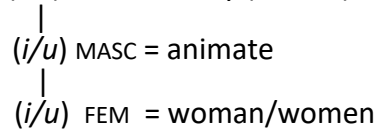
In Distributed Morphology, combinations of root and gender are regulated by licensing conditions (see Kramer 2015:50-57 among others), ensuring felicitous and interpretable output forms. Though Xhosa is quite flexible about the distribution of semantic fields across genders, individual nouns are limited to surfacing in particular pairs of classes. Licensing conditions are the theoretical means of encoding such restrictions. As an anonymous reviewer points out, stacking complicates them. For a noun like *umvundla* -- I return to this issue in §7.

## 5 Deriving the patterns

### 5.1 Xhosa conjoined singulars

I turn to the derivation of agreement with &P. Following A&A, features of conjuncts *percolate* to &P; resolved agreement is based on a *conversion* which intersects their *i*-features to form a single set of shared values. For A&A this is based on feature-dependencies among genders such as (73), for Greek. The lowest feature is the most specific; the most specific is always expounded.

(73) Greek gender in A&A:  $(i/u)$  CLASS = entity (neuter)

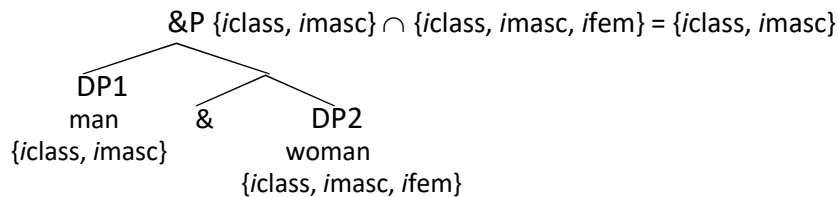


A&A derive agreement with the mismatched Greek conjuncts in (74) as shown (75). Their masculine and feminine features are interpretable because these are [animate] conjuncts.

Based on the dependencies in (73), (74) resolves to the most specific shared feature: masculine.

(74) O andras ke i gineka ine eksipni /\*eksipna. [Greek]  
 the.M.SG man and the.F.SG woman are intelligent.M.PL/ \*intelligent.N.PL  
 'The man and the woman are intelligent.'

(75) Deriving masculine agreement in the mismatched (74):



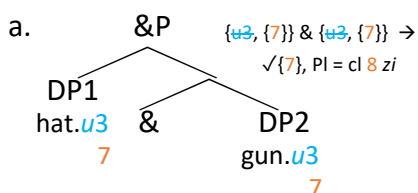
We do not find anything quite like the relationships in (73) among the features of Xhosa genders. The higher gender shells are not featurally more specific than the [human], [inanimate], [animal] genders that surface in agreement when conjuncts' genders mismatch or when nouns of classes 3/4 or 5/6 are conjoined. Nor is there evidence of nesting dependency relations among these three semantic cores in Xhosa, which seem instead to function as opposing and co-equal featural primitives. The approach that I advocate here in terms of Bantu *nP*-stacking is accordingly a bit different from A&A's, but their proposals of percolation and conversion provide important tools and insights, nonetheless.

As we proceed from this point it will be important to bear in mind that only formally identical noun class/gender features are intersected, such as [1&1], [7&7], etc; commonalities among "flavors" of mismatched genders are irrelevant. For example, *i*[entity] flavors of classes 1, 7, and 9 ensure that these genders always pattern as interpretable, unlike classes 3 and 5;

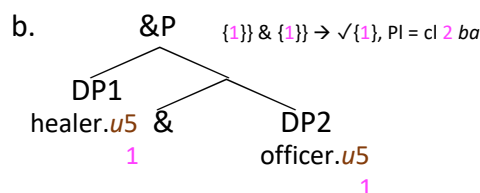
but the fact that they have the same minimal semantic content does not enable matching across genders – 1.*i*[entity] & 7.*i*[entity] are distinct genders, so cannot intersect. In other words, formal agreement reflects only formal gender features.

Building on §1 and §4 I propose that default 2, 8, and 10 agreement accompany conjoined singulars [3&3] and [5&5] because wholly *u*genders are ignored in the calculus of resolution, as (76)a,b illustrate. This can be modeled in terms of percolating feature bundles followed by deletion, exposing the inner gender D (7/8) core of [inanimate] nouns in (76)a = (38)a, or as *u*features failing to percolate as shown for (76)b = (38)b (henceforth features not labeled as *u*- are *i*-; examples reproduced below). For simplicity I adopt the non-percolation approach in (76)b (following A&A), and label the genders of singulars with singular class numbers (i.e. 1 rather than 1/2 or gender A).<sup>31</sup>

(76) Resolution in (38)a and (38)b via uniform percolation with deletion (a), or percolation only of *i*-features (b)



class 8 agr with conjoined inanimates [3+3]; = (38)a



class 2 agr with conjoined humans [5+5] = (38)b

(38) a. Um-nqwazi nom-pu **zi-se** tafile-ni.  
3-hat and.3-gun 8SM-are table-LOC  
'A hat and a gun are on the table.'

b. I-gqwetha ne-sela **√ba-phum-ile.**  
5-lawyer and.5-thief 2SM-leave-PST.DISJ  
'The lawyer and the thief left.'

In cases like (45)a and (47)b below, matching arbitrary flavors stack above matching *i*gender cores as shown in (77)a,b. Two proposals are relevant here: that arbitrary flavors are

<sup>31</sup> An anonymous reviewer asks if we could alternatively suppose that languages vary in whether *u*gender factors into resolution. A&A show that e.g. Greek [FEM<sub>arb.sg</sub> & FEM<sub>arb.sg</sub>] pair with gender-matching plural FEM.AGR unlike Xhosa [3&3] and [5&5]; under the reviewer's proposal perhaps Greek *u*FEM factors in while Xhosa 3 and 5 do not. But the genders of arbitrary Ns of Xhosa [1&1], [7&7], [9&9] *do* factor into resolved agreement like Greek [FEM<sub>arb.sg</sub> & FEM<sub>arb.sg</sub>] and unlike [3&3] and [5&5]. See discussion of (95)-(98).

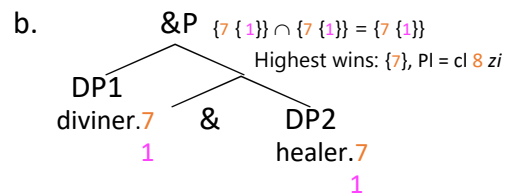
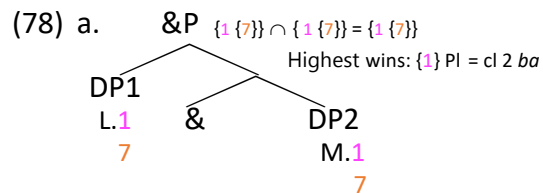
interpretable because they have the semantic content *i*[entity], and Kramer’s proposal for Somali gender-switch in (68) and (69) that when genders stack, the highest one wins. (77)a,b show that in (45)b and (47)b, the arbitrary genders are the highest for each nominal. Since both higher and lower gender features of each conjunct are interpretable, (78)a,b show that both *i*genders percolate from each DP to &P. I propose that their height asymmetry is preserved in the percolated feature sets as my bracketings in (78) indicate. As the same two *i*gender features are present on each conjunct, both are included in the intersection. Extending to resolution Kramer’s Highest-Wins proposal we predict resolved agreement in the formally matching arbitrary *i*entity flavors 1a and 9 for (45)b and (47)b respectively.

(45) a. U-L no-M **ba**-se tafile-ni. [1a&1a=2]  
 1a-L and.1a-M 2SM-LOC table-LOC  
 ‘The L and the M are on the table.’

(47) b. Isi-bhanxa nes-azi **zi**-ya-funda.  
 7-fool and.7-scholar 8SM- DISJ-study  
 ‘The fool and the scholar are studying.’

(77) a. [1a [7 train]] & [1a [7 machine]]

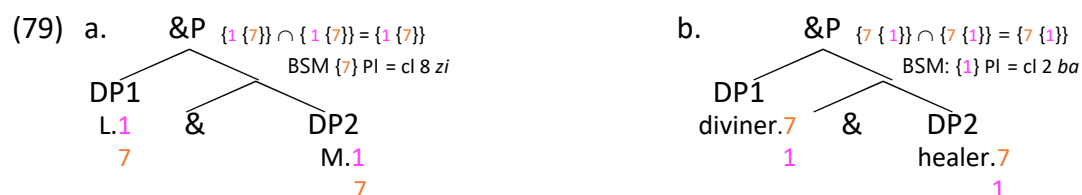
b. [7 [1 diviner]] & [7 [1 healer]]



This approach derives the majority agreement patterns for these examples, but as we have seen they are not the only licit choices. In cases like (45)a, speakers I consulted chose default *ba*- 23.33% of the time. And while 95% of Taraldsen et al’s speakers offered matching *zi*-agreement in (47)b, among speakers I worked with 30% offered only *ba*.

Such variation suggests that two slightly different grammars are involved in computing agreement with Xhosa conjoined subjects. I propose that in addition to the Highest-Wins grammar illustrated in (78), there is a second which I will call Best Semantic Match Wins (labeled

BSM in diagrams. I take this to mean that if two pairs of *igender* features in the intersection are total matches as in (45)b and (47)b, the most specific semantic match may take precedence as shown in (79)a,b. An *i*[entity] flavor in the intersection (one borne by arbitrary members of a gender) will generally lose out to a shared core of [human], [animal], or [inanimate] by this metric, as it does in a minority of cases as described in §3 and exemplified in (80)a,b.

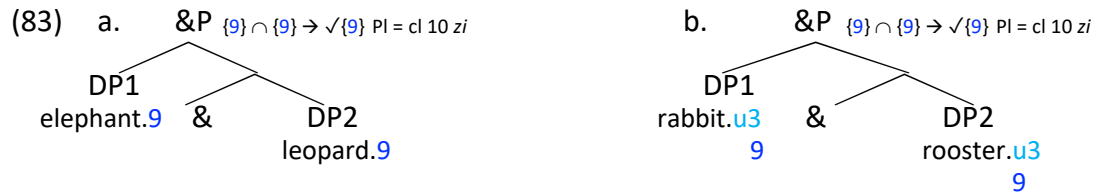


- (80) a. U-L no-M **zi-se** tafele-ni. b. Isi-bhanxa nes-azi **ba-ya-funda.**  
 1a-L and.1a-M 8SM-LOC table-LOC 7-fool and.7-scholar 8SM-DISJ-study  
 'The L and the M are on the table.' 'The fool and the scholar are studying.'

I propose that both grammars are available to speakers. As noted in §3, I encountered some variation in choices made by a single speaker across examples and meetings, suggesting that though there are general tendencies yielding the majority patterns, the choice is fairly free.

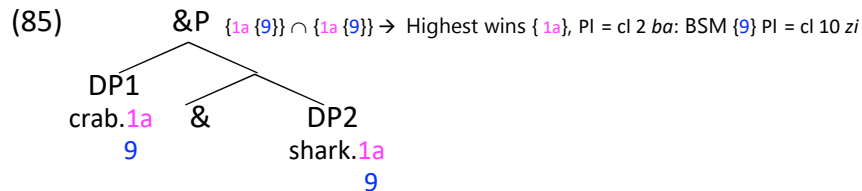
Turning to [animal] nouns, those of class 9 have single-layer *nP* structures shown in (81) and when conjoined, take class 10 *zi*-agreement as illustrated in (81) and (83)a. (83)b shows how gender mismatched [animal] conjuncts such as (82)a also combine with class 10 *zi*-agreement, as do animals of [5&5] and [3&3] (see (82)b). Familiarly, the *ugenders* do not factor into the intersection. Agreement is based on percolation of the class 9 cores.

- (81) In-dlovu (kanye) nen-gwe **zi-ya-lwa.** [9&9=10]  
 9-elephant and and.9-leopard 10SM-DISJ-fight.  
 'The elephant and the leopard are fighting.'
- (82) a. Isi-khova ne-hobe **zi-nokubhabha.**  
 7-owl and.5-dove 10SM-fly  
 'The owl and the dove can fly.'
- b. Um-vundla nom-qhagi **√zi-** /Xi-yanxila.  
 3-rabbit and.3-rooster 10SM /4SM-drunk  
 'The rabbit and the rooster are drunk.'



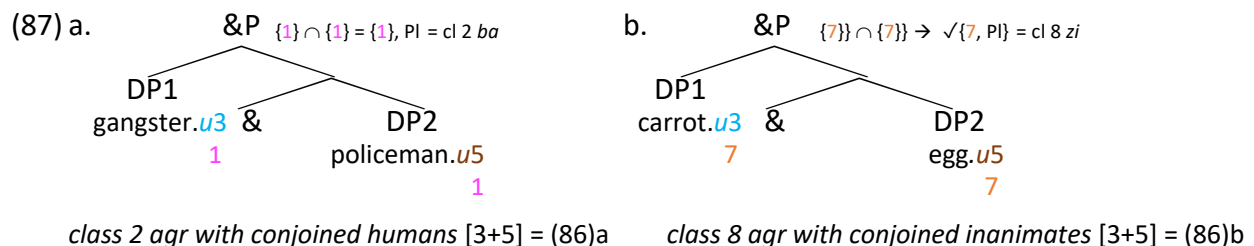
In (84), conjuncts of class 1a show us the distribution of matching vs. semantic agreement with [animal] nouns. We predict that in the Highest Wins grammar, the class 2 *ba*- plural for class 1a nouns will prevail. In the Best Semantic Match grammar, the class 10 plural for class 9 will result. Of 6 speakers I polled about this, 5 chose *ba*- and 1 *zi*-.

- (84) U-nonkala no-krebe **ba**-/zi-ya-lwa.  
 1a-crab and.1a-shark 2SM/10SM-DISJ-fight  
 'The crab and the shark are fighting.'



Examples (86)a,b involve [human] and [inanimate] nominals respectively, with mismatching *ugenders*. Underlying these genders in each instance are shared cores of class 1 and 7 respectively, as shown in (87)a,b. The *ugenders* are ignored; the cores value agreement.<sup>32</sup>

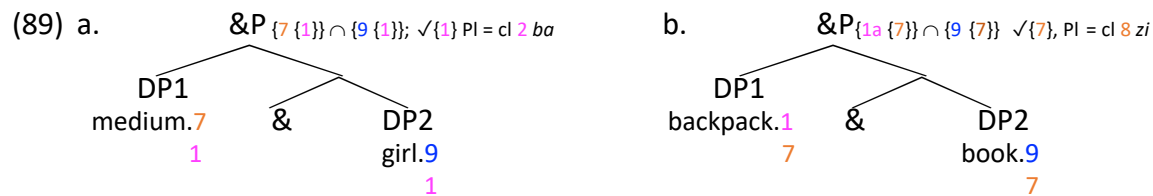
- (86) a. Um-gulukudu ne-polisa **ba**-sebenza ndawonye. [Xhosa]  
 3-gangster and.5-policeman 2SM-pres-work 9place.one  
 'The criminal and the policeman are working together.'  
 b. Um-nqathe ne-qanda **zi**-se tatile-ni. [Xhosa; Mitchely 2015:115]  
 3-carrot and.5-egg 8SM-loc table-LOC  
 'The carrot and the egg are on the table.'



<sup>32</sup> FCA is not impossible with mismatched conjuncts though was never spontaneously offered and was rejected by several speakers upon questioning about it. Default is the consistent choice.

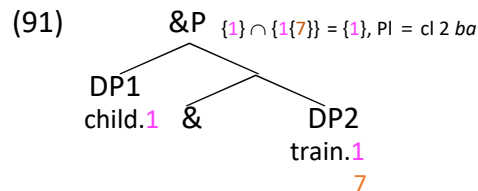
The higher genders are also mismatched in examples (88)a,b below. Since these are *igenders* they are not systematically excluded/deleted, but only formal gender features that are common to both conjuncts constitute the intersection. In both cases this is the core gender -- in (88)a class 1, as shown in (89)a, and in (88)b/(89)b, class 7.

- (88) a. Is-anuse nen-tombi **ba**-ya-sebenza. b. U-bhaka nen-cwadi **zi**-ngaphandle.  
 7-medium and.9-girl 2SM-DISJ-work 1a-backpack and.9-book 8SM-be.outside  
 'The medium and the girl are working.' 'The backpack and the book are outside.'



The analysis correctly predicts that class 1/2 can value agreement in (90)/(91), even though it is the [human] flavor on 'child' and the arbitrary [entity] flavor on 'train,' which has a distinct 7/8 *icore*. As the sole shared gender feature class 1 is the content of the intersection.

- (90) Um-twana no-lolilwe **ba**-ya-hamba  
 1-child and.1a-train 2SM-DISJ-move  
 'The child and the train are moving.'



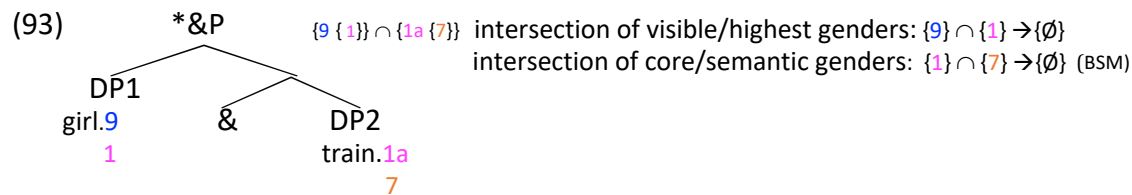
Agreement with conjoined humans and inanimates is generally ineffable if they are drawn from different genders, as also demonstrated by A&A for Greek. No choice of resolved agreement works in the following; a paraphrase like 'The girl collided with the train' is preferred.<sup>33</sup>

- (92) \*In-tombi no-lolilwe zi/ba-gilana.  
 9-girl and.1a-train 8SM-/2SM-collide  
 Intended: The girl and the train collided.

<sup>33</sup> As mentioned in §3.2.2 a couple of exceptions I encountered involved domestic animals like '1child and 9dog' which a few speakers accepted with class 2 agreement, matching the gender of 'child'.

The genders of the higher *ns* are mismatching {9&1a} and no shared *icore* exists, unlike in (88). And yet, both conjuncts do have class 1 features and *i*[entity] flavors of 9 and 1a.

Informally, it seems either visible genders or *icores* must match each other. (93) sketches how this works, for (92). The Highest Wins grammar fails since there is no match between highest class features 9 and 1a. Turning to the Best Semantics Match grammar, the class 1 flavors *i*[human] and *i*[entity] are not logically incompatible but they are not a match: this suggests that semantic matching must be total to prevail. The *i*[entity] features of class 9 ‘girl’ and class 1a ‘train’ cannot satisfy Best Semantics Match either, since they are in this case not associated with formally matching genders. A conspiracy of factors seems to rule resolved agreement out.



As for other pockets of semantic unity in genders, I attribute these to metalinguistic mapping principles like those in (94) rather than to a multiplicity of interpretable flavors within each gender.

- (94) Hypothetical Xhosa gender-mapping principles
- Language names → class 7
  - Alphabet letters → classes 1a/2a
  - Trees and long thin things → classes 3/4
  - Rounded things → classes 5/6
  - Fruit → classes 5/6
  - Stigmatized humans → 3/4
  - Borrowed days of the week → 3/4

It should be noted that my approach to uniform arbitrary conjuncts is a significant point of contrast with A&A. While they propose that *ugender* is excluded from the intersection, two instances of a single feature are treated as one in their account, creating a loophole that allows for feminine agreement in (95) rather than neuter, the Greek default for inanimates (see (96)).

(95) a. I fusta ke i bluza ine vromikes. [Greek; A&A:39]  
 the.F.SG skirt and the.F.SG t-shirt are dirty.F.PL  
 'The skirt and the t-shirt are dirty.'

b. [&P [skirt]<sub>UF.SG</sub> & [t-shirt]<sub>UF.SG</sub>]: no intersection to [neuter] since {fem} & {fem} = {fem}

(96) O pinakas ke i karekla ine vromika /\*vromiki/\*vromikes.  
 the.UM.SG blackboard and the.UF.SG chair are dirty.N.PL /dirty. M.PL/ dirty.F.PL  
 'The blackboard and the chair are dirty.'

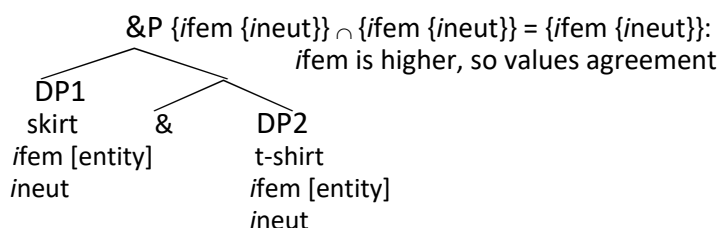
The approach in (95)b cannot be reconciled with the failures of matching [3&3], [5&5] or with the six Shona genders which pattern with them, as we saw in §3.5.

A&A also report that some Greek speakers accept neuter agreement with uniform masculine or feminine conjunctions of inanimates. Thus alongside of (95) some accept (97):

(97) %I fusta ke i bluza ine vromika. [Greek; A&A:42]  
 the.F.SG skirt and the. F.SG t-shirt are dirty.NEUT.PL  
 'The skirt and the t-shirt are dirty.'

The facts are compatible with the proposals that I made for Xhosa. Assuming Greek inanimates have neuter cores with associated [non-human] meaning, these conjuncts match in both their 'visible' feminine and their neuter *igender* core features; it is predicted that either Highest Wins or Best Semantic Match may determine agreement.

(98) Conjoined inanimates of matching arbitrary genders in Greek



## 5.2 Deriving the Shona patterns

Recall from §3.5 that there are only two options for agreement with conjoined singulars in

Shona: class 2 *va-* for [human] and class 8 *zi-* for non-human; see (63)b-d repeated below. Class

8 *zi-* is acceptable with class 7 conjoined humans as a secondary choice as shown in (62).

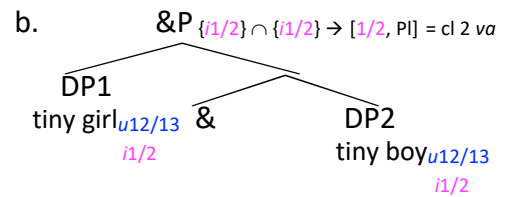
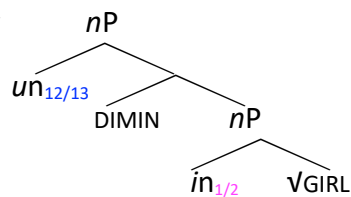
- (63) b. N-herera ne-n-yanzvi **va-** /\***dz**-aka-on-ana. [Shona: 9&9=2]  
 9-orphan and-9-expert 2<sub>SM</sub>-/\*10<sub>SM</sub>-PST-see-RECIP  
 'The orphan and the expert saw each other.'
- c. Im-bwa nhema ne-m-hou chena **zv-** /\***dz**-aka-rwa. [Shona: 9&9=8]  
 9-dog 9-black and-9-cow 9-white 8<sub>SM</sub>-/\*10<sub>SM</sub>-PST-fall  
 'The black dog and the white cow fell.'
- d. M-huno iyi ne nzeve iyi **zvi-** /\***dzi**-ne utachiona. [Shona: 9&9=8]  
 9-nose 9this and 9ear 9this 8<sub>SM</sub>-/\*10<sub>SM</sub>-have 14-infection  
 'This nose and this ear are infected.'
- (62) Chi-dhakwa ne-chi-komana **va-ri/?zvi-** ku-netsana. [Shona: 7&7=8 or 2]  
 7-drunkard and-7-small.boy 2<sub>SM</sub>/ 8<sub>SM</sub>-be 15-argue [human]  
 'The drunkard and the small boy argued.'

It seems clear that in Shona as in Xhosa, classes 1/2 include the *i*[human] flavor. As for 7/8, I propose that its semantic content is only *i*[entity]; it is the semantic core and default for all and only non-humans. The sole semantic cores underlying the Shona system are these two; the remaining 6 Shona genders have no association to classes of entities. They therefore pattern as purely uninterpretable and are excluded from the intersection in computations of resolved agreement in the same way as Xhosa classes 3 and 5.

The most striking and anomalous pattern in relation to this approach is that of diminutive formation, which clearly involves salient and predictable semantic content correlating with a special gender. But I suggest that *correlating* is the operative concept. While it is not unusual to assume causality, attributing the diminutive meaning to an interpretable gender/noun class feature of a particular *n* (Kramer 2015, Fuchs & van der Wal 2018), I instead assume following Carstens (1991, 2008) that the diminutive meaning belongs to a silent Bantu diminutive morpheme whose gender is uninterpretable (see also Creemers & Fenger 2018, Lowenstamm 2015, Simonović 2022, Wiltschko & Steriopolo 2007 for proposals that derivational affixes vary

in status and syntax).<sup>34</sup> Assuming the “little *n*” approach established in §4, a structural analysis is sketched in (99)a. There is no obstacle for *ka*- and *tu*- agreement with singleton diminutive DPs as in (66)b (see A & Van der Wal 2022:303 on agreement with derived nouns). But as we have seen, only interpretable features percolate and enter the intersection, determining resolved agreement. (99)b shows how default agreement derives for conjoined diminutives.

(99) a. structure for *kasikana*  
in (67)a



Summing up, a correlation of meaning and gender does not entail that the gender itself is semantic; as with any correlation, independent diagnostics are needed to establish causality. I propose that the availability of gender-matching agreement with uniform singular conjuncts provides this: it diagnoses gender interpretability while its impossibility indicates the opposite, in keeping with A&A’s proposal that resolved agreement reflects only *i*features.

## 6 A rejected alternative: semantic exceptionality yields failure

This section considers the most recent and detailed analysis of what property of BCS neuter gender makes it special, ruling out \*[neut&neut= neut.pl]. The Shona facts have argued that it is the matching pattern that is special; this section shows that diagnostic properties of neuter

<sup>34</sup> See also note 30 on prefix stacking in diminutives. An anonymous reviewer asks if there are nouns within Shona classes 12/13 that do not have diminutive meanings. This does not seem to be the case though other Bantu languages have noun classes that are dual-duty in this way; Fuchs & Van der Wal (2018) provide recent discussion and Luganda examples. The reviewer also questions the move in (99) to permit [DIMIN] in the ontology of features but not of gender features. This is a Shona-specific proposal, recognizing that semantics/gender relations are cross-linguistically idiosyncratic, including the fact that diminutives induce gender shift in some languages but not others (Wiltschko & Steriopolo 2007 among others). Under the analysis in (99) Shona [DIMIN] is an affix with semantic content rather like English *un*- and *re*- are, except that a licensing condition (on which see §4, §7, and Kramer 2015) always pairs it with a particular gender-bearing *n*. Another approach to the failure of matching agreement with diminutives might be that diminutive things are not a core entity type, in Shona; so an *n*[DIMIN] might be interpretable but its gender uninterpretable.

defectivity proposed in Arsenijević (2017, 2021) fail to accompany Xhosa classes 3 and 5.

Building on Arsenijević (2017, 2021), Alexiadou & Anagnostopoulou relate the BCS failure to the fact that many neuters are mass nouns, lacking individuation and number. This is taken to be intrinsic to the gender itself, affecting even nouns that are not themselves mass-denoting. As a result, BCS neuters pluralize irregularly if at all as exemplified in (100) for the noun ‘calf’. A suffix can make ‘calf’ into a collective feminine in (100)b; alternatively, a diminutive suffix yields a masculine pluralizable form in (100)c. What doesn’t work is a simple neuter plural, as illustrated in (100)d.

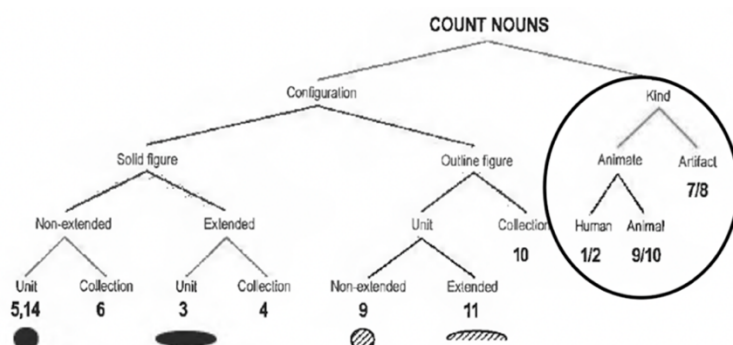
(100) Pluralization problem in BCS neuter gender

- |                                     |                                                          |                                                       |                                                             |
|-------------------------------------|----------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|
| a. tel-e<br>calf-NOM.N.SG<br>‘calf’ | b. tel-ad<br>calf-COLL.NOM.F.SG<br>‘calves’ (collective) | c. tel-ic-l<br>calf-DIM-NOM.M.PL<br>‘calves’ (plural) | d. *tel (-et)-a<br>calf-EXT-NOM.NEUT.PL<br>Intended: calves |
|-------------------------------------|----------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------|

There is a conceivable basis available for a similar kind of semantic differentiation among Bantu classes that do and do not pair with resolved agreement in a proposal of Denny & Creider (1986) that only proto-Bantu classes 1/2, 7/8, and 9/10 had semantic associations to entity types [human], [artifact], and [animal]; 3/4, 5/6 and a few others were instead characterized by abstract shape tendencies, in their view. I reproduce Denny & Creider’s schematic for this in Figure 1 where I circle their ‘kind’ node and its dependents. Denny & Creider argue that the number of reconstructed tree and plant terms in proto-Bantu classes 3/4 is statistically insufficient to justify their inclusion beneath this node (though this not uncontroversial; some later synchronic studies of individual languages provide support for the more traditional view. For example, Muluwa (2010) found a sizeable plurality of plant and tree terms, 46%, in classes

3/4 within five Bandundu languages of the Democratic Republic of the Congo).<sup>35</sup>

**Figure 1:** Proto-Bantu noun class semantics in Denny & Creider (1986)



A semantics of shape rather than entity type could conceivably be baked into classes 3/4 and 5/6 in the same way as mass is proposed for BCS neuter. Assuming entities are countable but properties like solid, outline, extended and the like are not, then the grammar of resolved agreement might be thwarted for [3&3] and [5&5]. However, we will now see that nouns of classes 3/4 and 5/6 are entirely regular according to a range of diagnostics that BCS neuters fail.

If semantic content underlies the failure of matching agreement with [3&3] and [5&5], there is no evidence that this relates to individuation or a lack of countability. Nouns of these classes are mostly ordinary objects that are countable and pluralize regularly as shown in (101).

- (101) a. **um**-thi **omu**-nye    b. **imi**-thi **imi**-thathu    c. **i**-qanda **eli**-nye    d. **ama**-qanda **ama**-thathu  
       3-tree 3-one            4-trees 4-three            5-egg 5-one            6-eggs 6-three  
       ‘one tree’            ‘three trees’            ‘one egg’            ‘three eggs’

Apart from class 1, Xhosa classes typically include some noun or nouns referring to uncountable substances such as liquids and/or masses as I showed in (32), repeated below.

<sup>35</sup> For Xhosa, Johnson (1990) lists 284 tree names of which 66.67% are class 3/4; the remaining 33% are scattered across classes 5/6, 7/8, and 9/10. Bhatt (2013) lists 30 plants and trees of which 8 are in classes 3/4 = 26%; Bhatt (2014) lists 11/29 in class 3/4 -- 61%. In Wehmeyer & Rose (1983) plants and trees in classes 3/4 number 3/18 or 17%. The smaller % within 3/4 in the latter three studies seems to correlate with their focus on medicinal plants, whose names are more scattered. Unfortunately, most technical species names for plants and trees in these studies are unfamiliar to the Xhosa university students who serve as my consultants.

(32) Liquids and masses: scattered

- |                                           |                                                |                                          |                                                 |                                     |                                        |
|-------------------------------------------|------------------------------------------------|------------------------------------------|-------------------------------------------------|-------------------------------------|----------------------------------------|
| a. <b>um</b> -chamo<br>3-urine<br>'urine' | b. <b>i</b> -gazi<br>5-blood<br>'blood'        | c. <b>ama</b> -nzi<br>6-water<br>'water' | d. <b>isi</b> -dudu<br>7-porridge<br>'porridge' | e. <b>i</b> -oili<br>9-oil<br>'oil' | f. <b>u</b> -bisi<br>11-milk<br>'milk' |
| g. <b>um</b> -gubo<br>3-flour<br>'flour'  | h. <b>ama</b> -zimba<br>6-sorghum<br>'sorghum' | i. <b>i</b> -tyuwa<br>9-salt<br>'salt'   | j. <b>u</b> -daka<br>11-mud<br>'mud'            |                                     |                                        |

These nouns must combine with a unit counter for enumeration, as shown in (102)a. But as (102)b indicates, my consultants offered as the counter for liquids *ithonzi* - 'drop' in (102)b, itself a noun of outlier class 5. Furthermore (102)b and (103)a show that a unit counter is required for a non-count noun in class 7 – a class in which conjoined singulars may take regular plural agreement (see (103)b,c). Taken together this and the examples in (101) argue strongly against any deficiency in countability correlating with the need for default agreement with conjoined singulars of classes 3 and 5.

- (102) a. #i-gazi eli-nye      b. i-thonzi (eli-nye) l-e-gazi /l-esi-dudu  
5-blood 5-one      5-drop 5-one 5-of.5-blood/5-of.7-porridge  
Literally: 'one blood'      'a/one drop of blood/porridge' [unit counter is cl.5]
- (103) a. #isi-dudu esi-nye      b. isi-tya esi-nye      c. izi-tya ezi-mbini  
7-porridge 7-one      7-dish 7-one      8-dish 8-two  
Literally: 'one porridge'      'one dish'      'two dishes'

Arsenijević (2017, 2021) provides an additional diagnostic for defectiveness of BCS nouns in neuter gender: he reports that neuter is unavailable in "highly individuated plural uses such as....in demonstratives with pointing..." ((104) from Arsenijević 2021:21). (105)a-d show that this is not a problem for classes 5/6 and 3/4, whether singular or plural.

- (104) Ona      \*(deca)      su      neumorna.      (pointing at children)  
those.NEUT.PL child.NEUT.PL AUX.PL tireless.NEUT.PL  
'Those children are tireless'/'\*Those are tireless.'
- (105) a. **Loo** (m-thi /m-gulukudu) m-de.      b. **Eli** (qanda/gqwetha) l-awa.  
3.that 3-tree /3-gangster 3-tall      5.that 5egg/5lawyer 5SM-fell  
'That (tree/gangster) is tall'      'That (egg/lawyer) fell'
- c. **Le** (mi-thi /mi-gulukudu) im-bi.      d. **La** (ma-qanda/ma-nenekazi) ma-hle.  
4.those (4-trees/4-gangsters) 4-bad      6.those(6-eggs/6-ladies) 6-nice  
'Those (trees/gangsters) are bad.'      'Those (eggs/ladies) are nice.'

Another area of potential contrast between the outlier and well-behaved genders of Xhosa is pronominal binding. In some languages, where a noun's grammatical gender is at odds with its semantic gender, a bound pronoun may match the latter rather than the former. Example (106) (from Spathas 2010:222) shows that a semantically feminine but formally neuter noun in Greek can bind either a neuter or feminine pronoun. (107) shows that in contrast, bound readings for Xhosa pronouns require matching, in every class.<sup>36</sup> (107)a-c have [human] subjects of arbitrary genders binding possessive pronouns; the pronouns must match the subject that binds them in class (though the connector *-a* agrees in noun class with the possessed). The subject in (107)d is an [inanimate] of arbitrary gender class 1/2. The possessive pronouns it binds cannot be of any class other than class 1.

- (106) Kathe koritsi diakosmise to dhomatio {tis/tu} [Greek]  
 every girl.N decorated the room hers/its  
 'Every girl decorated her room.'
- (107) a. I-nenekazi nga-li-nye<sub>i</sub> I-a-yi-funda in-cwadi ya-lo<sub>i</sub> /ya-khe<sub>j</sub>. \*i = j  
 5-lady each-5-one 5SM-PST-9OM 9-book 9-5POSS /9-1POSS  
 'Each lady<sub>i</sub> read her<sub>i</sub> book.'
- b. Y-onke in-kwenkwe<sub>i</sub> y-a-tya imi-funo ya-yo<sub>i</sub> /ya-khe<sub>j</sub>. \*i = j  
 9-every 9-boy 9SM-PST-eat 4-vegetables 4-9POSS /4-1POSS  
 'Every boy ate his vegetables.'
- c. Y-onke imi-gulukudu<sub>i</sub> y-a-khwela ama-hashe a-yo<sub>i</sub> /a-bo<sub>j</sub>. \*i = j  
 4-all 4-gangster 4SM-PST-ride 6-horses 6-4POSS /6-2POSS  
 'All the gangsters rode their horses.'
- d. U-lolilwe nga-m-nye<sub>i</sub> w-a-phulukana ne-vili la-khe<sub>i</sub> /la-so<sub>j</sub>. \*i = j  
 1a-train each-1-one 1SM-PST-break PREP-5-wheel 5-1POSS /5-7POSS  
 'Every train broke its wheel'

Summing up, unlike BCS neuter, Xhosa genders 3/4 and 5/6 display no systematic

<sup>36</sup> An anonymous reviewer points out that whether pronouns match the semantic or arbitrary gender of their antecedents and binders in a language is logically independent of the properties of specific genders. I include the results since if there *were* contrasts in this respect between classes 3/4 and 5/6 on the one hand and 1/2, 7/8, 9/10 on the other, the correlation with agreement asymmetries would be analytically significant.

irregularities correlating with the failure of matching agreement with [3+3], [5+5].<sup>37</sup> Sensitivity to individuation and countability seem conceptually relevant to the calculus of agreement with conjuncts, but do not help us define Xhosa's outliers. While space considerations preclude my replicating the same tests for Shona, the results are exactly the same.

## 7 Conclusions

I have argued that a modest but grammatically significant semantic core underlies the Bantu noun class system: in Xhosa [human], [inanimate], and [animal]; in Shona [human] vs [entity]. So-called default or semantic agreement with mismatched conjoined singulars in Bantu languages is actually syntactic agreement with one of these *i*gender cores.

Better-studied languages with two or three genders give the impression that gender-matching plural agreement with conjoined uniform singulars is the norm and its failure an aberration to be explained. The Bantu languages of this study, particularly Shona, turn that on its head, arguing that genders in which matching agreement succeeds are the special ones.

I've argued that associations to broad and salient classes of entities characterize the genders in which gender-matching plural agreement succeeds, with uniform singular conjuncts; conceptually motivated and arbitrary members of a given gender tend to pattern alike in this

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<sup>37</sup> Mletshe (2019) claims classes 3/4 are not targets of nominalization in the way the other classes are, yet there are very many 3/4 deverbal nouns including *um/imiculo* – 'music/song' < *cula* – 'to sing,' *um/imidlalo* 'game' < *dlala* – 'to play,' *um/imibuzo* 'question' < *kubuza* – 'to question,' *um/imifanekiso* – 'picture' < *kufana* – 'to resemble,' *um/imiboniso* – 'exhibition/performance' < *kubonisa* – 'cause to see.' Since Mletshe was concerned with [human] nominalizations, his results might reduce to 3/4's inhospitableness to [human] Ns. See also note 11 on final vowel nominalizers, which may have their own gender features. A reader asked if classes 3/4 and 5/6 accept borrowings; they do; e.g. borrowed English days of the week are class 3 nouns and among 5/6 borrowings are compounds involving paper of the form *iphepha+X* such as *iphepha.mvume* – 'permit,' *iphepha.ndaba* – 'newspaper.' Futuse (2019) notes that class 5 loan words include miscellany (ticket, police, wheel, cloth) and terms for fruit, though 81.5% of borrowings appear in 9/10; Borrowings into classes 7/8 apparently begin with [s] as this is compatible with the [isi-] prefix of this class. What semantic mappings there are of borrowings to genders may follow from principles like (94).

respect (though semantic agreement is the choice of some speakers, in some examples). The pattern motivates my proposal that genders with any interpretable content typically have an *i*[entity] flavor that combines with arbitrary members; this enters the intersection on &P, enabling matching agreement.

Under this view, genders in which uniform conjoined singulars systematically fail to pair with gender-matching plural agreement are genders without *i*[entity] at all. This is a proposal with consequences for the theory of gender in general.

As for BCS neuter, it might be that there is more than one source for failures of resolved agreement, the absence of countability being a feature that thwarts it for this particular gender as Arsenijević (2017, 2021) and Alexiadou & Anagnostopoulou (2024) propose. Since Shona classes 9/10 are largely composed of countable contents and resolved agreement fails nonetheless, this alone cannot be criterial. A unified account may be possible, assuming that as a general rule, to pattern as interpretable and thus succeed in the calculus of resolved agreement, any gender must have semantic association to a salient class of *countable* entities. This is true of entities that are feminine, masculine, human, inanimate, and animal – the semantic cores of genders we have seen pattern together. If, on the other hand, a gender is purely formal, with no associated semantic core, it does not matter what its individual contents are, in the approach I have advocated. It will pattern as uninterpretable, and resolved agreement will fail, as I posit to be true of Shona 9/10, and 3/4 and 5/6 in both Bantu languages.

Turning to the mechanics of the account, I have adopted much of a percolation and conversion approach proposed in A&A for agreement resolution for conjoined singulars and

concur with their proposal that default/semantic agreement is actually syntactic agreement. One important aspect of their system does not play a role in my account of Xhosa and Shona: feature hierarchies and the dependency relations within them. Instead, I have argued that the features of so-called default agreement are contributed by core *n*Ps encoding three opposing semantic primitives. Bantu's resolved agreement is not determined by working from most to least specific along a feature hierarchy as in the languages of A&A's study, because Bantu genders are organized along different lines. That such differences exist is a significant finding for the study of gender.

§4 deferred discussion of a significant theoretical consideration: licensing conditions, which in Distributed Morphology are hypothesized to regulate combinations of *ns* and roots. My proposals entail that many Xhosa roots must combine with two *ns*:  $\sqrt{\text{MNTU}}$  of (10)a (repeated below) combines only with *in* of classes 1/2, but  $\sqrt{\text{GQWETHA}}$  of (10)b has the two-layered structure illustrated, combining first with *in* of classes 1/2 and subsequently with *un* of classes 5/6. Under the approach I have advocated, certain *ns* always combine with core *n*Ps. Assuming roots head-adjoin to *ns*, an outer *n* like  $un_{5/6}$  in (10)b has simultaneous access to the lower  $in_{1/2}$  and the root it has combined with; this should suffice to ensure that a licensing condition can apply after  $un_{5/6}$  is added, finding  $[un_{5/6} [in_{1/2} \sqrt{\text{GQWETHA}}]]$  to be well-formed.

(10) Sample structures of [human] nouns: a core of interpretably [human] 1/2

- |    |                                                            |                                                 |                                   |
|----|------------------------------------------------------------|-------------------------------------------------|-----------------------------------|
| a. | um-ntu/aba-ntu<br>1-person/2-people<br>'person/s'          | [ $in_{1/2} \sqrt{\text{MNTU}}$ ]               | $in_{1/2}$ for [human]            |
| b. | i-gqwetha /ama-gqwetha<br>5-lawyer/6-lawyers<br>'lawyer/s' | [ $un_{5/6} [in_{1/2} \sqrt{\text{GQWETHA}}]$ ] | $un$ of 5/6 stacks above 1/2 $in$ |

But what of the inner layer, at the point when  $in_{1/2}$  combines with the root? Kramer (2015) proposes that licensing conditions for interpretable gender features apply at LF, while those for  $ugender$  features apply at PF. Under this view, if the outer layer gender of a Xhosa noun is uninterpretable as is true of the class 5/6 *igqwetha/amagqwetha* – ‘lawyer/s’, then the inner and outer layers differ in the locus of scrutiny. This entails that  $[in_{1/2} \sqrt{GQWETHA}]$  must be evaluated without reference to the outer layer. Yet this root cannot licitly surface as a 1/2 noun: *\*umgqwetha/abagqwetha*. It appears that the scope of the relevant licensing condition is to narrowly ensure that the  $n$ +root combination is licit, not that it is viable to surface as it stands. This is not implausible, given that  $n$ +root typically does not surface bare; in addition to any outer  $nP$  layers there is inflection to be added and assessed for well-formedness.

Thus the LF licensing condition will find  $[in_{1/2} \sqrt{GQWETHA}]$  licit, and  $un_{5/6}$  will select it, having access to the full  $n$ +root combination, and this will in turn be inspected. The result is a complex array of licensing conditions, but this seems justified by the idiosyncrasies and semantic unpredictability of Xhosa noun class assignments. Many questions no doubt remain which I leave to future research.

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